

Fit Is Not Enough: An Empirical Study of eSSVI Calibrations Through the Local Volatility They Induce

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Abstract

A smile parameterization is selected on in-sample fit and then used for something the fit never tested: the Dupire local volatility that prices the exotic book. On real option chains, that gap is where models fail. Every capture in our panel (two weeks of delayed CBOE quotes on four index ETFs and four single names, sampled every fifteen minutes) arrives with static-arbitrage violations in its raw quotes, American early-exercise premia that bias the parity forward, and expiries dense enough that every interpolation seam reaches the local volatility.

This paper measures how eSSVI calibrations survive that environment. One pipeline de-Americanizes every quote with a measured carry, repairs static arbitrage once and identically for every model, and follows six calibrations through fit, out-of-sample interpolation, admissibility, induced local volatility, and knock-out prices on shared Monte Carlo paths: the sequential penalty-based eSSVI of Hendriks–Martini; the global parametrization of Mingone (2022), arbitrage-free by construction at its fitted expiries, with the backbone exponent both pinned at $\gamma = \frac{1}{2}$ and fitted; a new greedy variant that delivers the same guarantee at per-expiry speed; SSVI; and raw SVI as the unstructured benchmark.

Three results matter for a desk. In-sample fit misleads: held out of sample, the best-fitting benchmark loses nearly half its advantage while the structured calibrations generalize at their in-sample error. The construction-level guarantee holds exactly where penalties leak, and the greedy pass delivers it in milliseconds. The sharpest comparison the design permits is between that greedy pass and the joint solve at the same exponent: two arms sharing slice family, backbone, guarantee, fitted pillars and Monte Carlo paths, differing only in the algorithm that locates their parameters. They price a one-month knock-out 3.8% apart, a wider gap than enforcing the guarantee produces at all, so calibration mode is a pricing decision rather than an implementation detail. The jointly calibrated surfaces also carry a systematic at-the-money bias between expiries. The exponent pin is nearly immaterial in-sample and costs only at the long end out of sample. Pipeline, chains, and every number are released.

Keywords: implied volatility surface; eSSVI; SSVI; no-arbitrage calibration; Dupire local volatility; barrier options; calibration speed; out-of-sample validation; equity options.

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1 Introduction

The implied volatility surface is the market’s coordinate system: quoting, hedging, structuring, and every smile-consistent model downstream (Dupire local volatility, stochastic-local-volatility leverage functions, variance-swap replication) start from it. The surface itself is not observed. What the market provides is a discrete, noisy, occasionally arbitrageable set of bid/ask quotes on a strike–expiry grid, and turning those quotes into a surface means choosing a *smoothing parameterization*. In practice that choice is defended on a single number, the in-sample fit error.

In-sample fit is not a sufficient criterion. A surface is not used where it is fitted; it is used downstream, by the local-volatility model that prices the exotic book, after Dupire’s formula has divided a time derivative of total variance by a quantity containing its second strike derivative. Structure imposed across expiries is therefore consequential at the level of the local volatility: it is differentiated once in t and twice in k before it reaches a price. Two surfaces that agree on quoted vols to a few basis points can induce local volatilities that differ by an order of magnitude in curvature, and a barrier option, whose value depends on σ_{loc} in the neighbourhood of the barrier, turns that difference into a price. How well a parameterization fits the quotes is therefore weak evidence of how it behaves downstream.

This paper therefore evaluates smile parameterizations where their consequences are realized. Six calibrations are fitted to identical, cleaned, de-Americanized chains from a delayed retail-accessible feed, and each is followed through the same chain:

parameterization $\rightarrow$ fitted parameters, their stability and their admissibility $\rightarrow$ Dupire local volatility $\rightarrow$ barrier price.

Every stage is a measurement. In-sample fit is recorded and then tested out of sample with a leave-one-expiry-out experiment, which separates structure from flexibility. Admissibility is measured on a dense grid rather than assumed from the construction, the induced local volatility is scored for roughness and containment, and the barrier prices are computed on shared Monte Carlo paths so that only the surface differs, read against a noise floor set by refitting each calibration on the next snapshot.

The six arms. We evaluate three parameterizations across six distinct calibration methods, set out in Table 1. Three are eSSVI slices (Hendriks and Martini, 2019) carrying the Hendriks–Martini no-arbitrage conditions *by construction*: two produced by the global recursion of Mingone (2022), with the power-law backbone exponent pinned at the theoretically distinguished value $\gamma = \frac{1}{2}$ and fitted freely (G-eSSVI), and one produced by a greedy forward pass developed here, which keeps each slice inside its predecessor’s admissible region at per-expiry speed with a single free parameter per slice (S-eSSVI, Section 3.4). Against them stand the Hendriks–Martini calibration as practised, with slices fitted in time order under penalties against calendar crossing and butterfly violation; SSVI (Gatheral and Jacquier, 2014), the fully global comparison point, whose no-arbitrage conditions are sufficient only while its power-law exponent stays in $(0, \frac{1}{2}]$; and raw SVI (Gatheral, 2004), five parameters per expiry with no cross-expiry structure. SVI is not a candidate for production use. It serves as the *unstructured benchmark*, the endpoint of the structure axis against which the others are read.

We call the three arms whose no-arbitrage conditions hold by construction the *guaranteed arms*. The guarantee covers the *fitted pillar slices* and does not automatically survive the interpolation between expiries, so the between-pillar surface is measured on a dense grid rather

	S-eSSVI $\gamma=\frac{1}{2}$	G-eSSVI $\gamma=\frac{1}{2}$	G-eSSVI γ free	eSSVI (seq.)	SSVI	SVI (bench.)
Free parameters	1/expiry + 1 global	$2N+1$ global	$2N+2$ global	2/expiry	3 global	5/expiry
Calibration	greedy	joint	joint	sequential	joint	per expiry
Butterfly at pillars	by constr.	by constr.	by constr.	penalized	cond. $\gamma \leq \frac{1}{2}$	none
Calendar at pillars	by constr.	by constr.	by constr.	penalized	by constr.	none
Skew term structure	power law	power law	power law	free, twists	one global correlation	free/expiry
Interpolation stays in family	yes	yes	yes	yes	yes	no
Closed-form butterfly / Dupire	yes	yes	yes	yes	yes	yes
ATM level	quoted	joint fit	joint fit	quoted	quoted	by-product

Table 1: The six arms. “By constr.” marks conditions that hold for any admissible parameter vector — Mingone’s Proposition 3.1 for the joint global arms, the pairwise greedy-tube argument of Section 3.4 for S-eSSVI; “penalized” marks conditions encouraged by a calibration penalty and measured afterwards; SSVI’s butterfly guarantee is conditional on the fitted exponent staying in $(0, \frac{1}{2}]$, which Section 7.1.4 shows it does not. N is the number of fitted expiries. The parameter budgets differ by construction (that is the design variable), so in-sample fit is an *input* to the chain, not a verdict.

than assumed clean (Sections 4 and 7.1.4). “Guaranteed” below always means guaranteed at the pillars.

The paper’s subject is the comparison of the guaranteed arms with the penalty calibration and with each other, and one contrast is unusually clean. S-eSSVI and G-eSSVI at $\gamma = \frac{1}{2}$ share the slice family, the power-law backbone, the exponent, the pillar guarantee, the repaired quotes they are fitted to and the Monte Carlo paths they are priced on. They differ in one thing: whether the parameters are located by a greedy forward pass or by a joint solve. Any difference measured downstream is therefore attributable to the calibration algorithm alone, which is a variable usually chosen on speed and treated as an implementation detail. The two G-eSSVI arms isolate the γ pin the same way, and guaranteed-versus-penalty isolates enforcement. SSVI locates the family against the fully global alternative, and SVI calibrates how much fit an unstructured smile can extract from the same quotes.

We therefore evaluate not only fit and arbitrage, but also local-volatility roughness and downstream option prices. Barrier pricing is used only as a downstream sensitivity test; monitoring, correction and barrier conventions are held fixed across calibrations.

Contributions.

1. **An end-to-end empirical comparison.** Six calibrations are followed on one open pipeline (RustyQLib) from fit through out-of-sample interpolation, admissibility, induced local volatility and exotic prices. Every chain is de-Americanized and arbitrage-repaired once, and all six arms share the cleaning, inversion, Dupire construction, solver and Monte Carlo draws, so measured differences are properties of the calibrations rather than of the treatment.
2. **A methodological contribution.** An open implementation of the Mingone (2022) global eSSVI construction, mechanically verified, and S-eSSVI, a greedy one-parameter-per-slice variant that carries the same pillar guarantee at per-expiry cost (Section 3.4).
3. **Empirical findings.** Fit, admissibility, local-volatility behaviour and exotic prices rank

the calibrations differently. The best-fitting arm generalizes worst out of sample; round-trip repricing, the diagnostic practitioners commonly reach for, does not separate the arms; and calibration mode moves the barrier price more than the enforcement choice itself.

Related work. The construction literature is mature: [Homescu \(2011\)](#) surveys methodologies, [Fengler \(2009\)](#) smooths with arbitrage-free splines, [Roper \(2010\)](#) gives the static-arbitrage conditions used below, and [Gatheral and Jacquier \(2014\)](#) introduce SSVI together with the butterfly bounds we enforce. [Hendriks and Martini \(2019\)](#) introduce eSSVI; [Corbetta et al. \(2019\)](#) and [Martini and Mingone \(2022\)](#) develop the analytic no-arbitrage domain for its slices; [Mingone \(2022\)](#) contributes the global parametrization this paper implements, in which the no-arbitrage conditions hold by construction and all maturities are calibrated jointly. [Cont and da Fonseca \(2002\)](#) study the dynamics of fitted surfaces, which is the frame for our stability instrument.

2 Setting and no-arbitrage preliminaries

2.1 Notation

Fix a valuation date t_0 and an underlying with forward curve F_t (construction in Section 5). For an option quote with strike K and expiry t , write the Black–Scholes implied volatility $\sigma_{\text{BS}}(K, t)$, the *forward log-moneyness*

$$k = \ln \frac{K}{F_t}, \quad (1)$$

and the *total implied variance*

$$w(k, t) = \sigma_{\text{BS}}^2(F_t e^k, t) t. \quad (2)$$

All the parameterizations are naturally expressed in (k, w) coordinates, and we compare them there. That k is measured against the *forward* rather than against spot is not a notational detail: it fixes the drift under which the induced local volatility is the correct one (Section 2.3).

2.2 Static arbitrage in (k, w) coordinates

Following [Roper \(2010\)](#) and [Gatheral and Jacquier \(2014\)](#), a surface $w(k, t)$ that is (i) free of *butterfly* arbitrage in every slice and (ii) free of *calendar* arbitrage across slices, and satisfies mild limit conditions, defines arbitrage-free call prices.

Definition 1 (Butterfly condition). *For a smile $w(\cdot)$ at fixed expiry, define*

$$g(k) = \left(1 - \frac{k w'(k)}{2 w(k)}\right)^2 - \frac{w'(k)^2}{4} \left(\frac{1}{w(k)} + \frac{1}{4}\right) + \frac{w''(k)}{2}. \quad (3)$$

The slice is free of butterfly arbitrage iff $g(k) \geq 0$ for all k (and call prices tend to zero as $k \rightarrow \infty$). The risk-neutral density is recovered as

$$p(k) = \frac{g(k)}{\sqrt{2\pi w(k)}} \exp\left(-\frac{1}{2}d_-(k)^2\right), \quad d_-(k) = -\frac{k}{\sqrt{w(k)}} - \frac{\sqrt{w(k)}}{2}, \quad (4)$$

so $g < 0$ is exactly a negative density ([Breedon and Litzenberger, 1978](#); [Gatheral and Jacquier, 2014](#)).

Definition 2 (Calendar condition). *The surface is free of calendar arbitrage iff $w(k, t)$ is non-decreasing in t at every fixed k .*

Two classical constraints recur below. First, [Lee \(2004\)](#): an arbitrage-free smile has asymptotically linear total variance with wing slopes at most 2,

$$\limsup_{|k| \rightarrow \infty} \frac{w(k)}{|k|} \leq 2. \quad (5)$$

Second, Dupire’s local variance in Gatheral’s form: if the surface is smooth,

$$\sigma_{\text{loc}}^2(k, t) = \frac{\partial_t w(k, t)}{g(k, t)}, \quad (6)$$

so butterfly violations ($g \leq 0$) and calendar violations ($\partial_t w < 0$) are precisely the failure modes that make a fitted surface unusable for local-volatility pricing. Equation (6) is the hinge of this paper: it is why the roughness link and the admissibility link of the chain are the same link seen twice, once as a magnitude and once as a count of violations.

2.3 Local volatility and the drift it implies

Dupire’s local volatility ([Dupire, 1994](#); [Derman and Kani, 1994](#)) is the unique one-factor Markovian diffusion that reprices an entire arbitrage-free vanilla surface. Written in the forward log-moneyness coordinate used above, (6) defines σ_{loc} for the dynamics

$$\frac{dS_t}{S_t} = \frac{d \ln F(t)}{dt} dt + \sigma_{\text{loc}}(S_t, t) dW_t, \quad (7)$$

that is, with the drift set by the *chain’s own* forward curve and not by the risk-free rate. For a dividend payer, or a name with borrow, the two differ materially. The distinction is not a second-order refinement but the principal source of replication error in this study, and [Section 7.2.2](#) returns to it with measurements.

Two consequences drive the design.

First, (6) is a *second-order* functional of the surface: it divides $\partial_t w$ by g , which itself contains w'' . Smoothing error is therefore differentiated twice before it reaches the price, and features invisible in a fitted-vol comparison — a small curvature error, a kink in the term structure at a pillar expiry — can dominate the local volatility.

Second, the model carries its own acceptance tests. Because (7) is *constructed* to reproduce the calibrating surface, any repricing error is calibration or numerical error rather than model error, so round-trip repricing is unambiguous as a criterion — and, as it turns out, uninformative as a discriminator ([Section 7.2.1](#)). The spot must likewise satisfy $\mathbb{E}[S_T] = F_T$ under (7), which fails when σ_{loc} is clamped or extrapolated badly in the wings and probability mass leaks.

3 Three smile parameterizations

3.1 Raw SVI

The raw SVI parameterization ([Gatheral, 2004](#)) of one expiry’s total variance is

$$w_{\text{SVI}}(k) = a + b \left[\rho(k - m) + \sqrt{(k - m)^2 + \sigma^2} \right], \quad (8)$$

with level a , wing slope $b \geq 0$, skew $\rho \in (-1, 1)$, shift m , and ATM curvature $\sigma > 0$; admissibility additionally requires the minimum total variance $a + b\sigma\sqrt{1 - \rho^2} \geq 0$. The wings are asymptotically linear,

$$w_{\text{SVI}}(k) \sim b(1 \pm \rho)|k| \quad (k \rightarrow \pm\infty), \quad (9)$$

so Lee’s bound (5) is respected whenever $b(1 + |\rho|) \leq 2$. SVI’s decisive practical advantage is that (8) is C^∞ with *closed-form* derivatives: $g(k)$ in (3) and the Dupire numerator and denominator in (6) are analytic, which makes the induced local volatility smooth by construction *within* a slice. Butterfly arbitrage is *not* guaranteed — the Vogt example reproduced in our test suite carries $\min_k g \approx -0.033$ — so $g(k)$ must be checked per fitted slice (Gatheral and Jacquier, 2014).

Five parameters per expiry is the largest budget among our contenders: on an eleven-strike single-name slice SVI spends nearly half the degrees of freedom in the data, which is exactly the flexibility-versus-noise tension this paper measures. The parameters are also weakly identified when the smile is close to symmetric, (a, b, σ) trading off against one another; that is the documented instability discussed by Zeliade Systems (2009) and one of the motivations for SSVI.

3.2 SSVI

SSVI (Gatheral and Jacquier, 2014) parameterizes the whole surface from the ATM total-variance term structure $\theta_t = w(0, t)$ and a curvature function φ :

$$w_{\text{SSVI}}(k, t) = \frac{\theta_t}{2} \left[1 + \rho \varphi(\theta_t) k + \sqrt{(\varphi(\theta_t)k + \rho)^2 + 1 - \rho^2} \right], \quad (10)$$

with the power-law curvature used throughout this paper,

$$\varphi(\theta) = \frac{\eta}{\theta^\gamma (1 + \theta)^{1-\gamma}}, \quad \gamma \in (0, 1], \eta > 0. \quad (11)$$

Three global parameters (ρ, η, γ) shape every expiry; the term structure enters only through θ_t , taken from the quoted ATM variances (interpolated linearly between pillars, proportionally below the first). Its selling point is Theorem 4.1–4.2 of Gatheral and Jacquier (2014): with θ_t non-decreasing (calendar) and, at each pillar,

$$\theta\varphi(\theta)(1 + |\rho|) \leq 4, \quad \theta\varphi(\theta)^2(1 + |\rho|) \leq 4, \quad (12)$$

the surface is free of static arbitrage. For the power law (11) with $\gamma \in (0, \frac{1}{2}]$, the condition $\eta(1 + |\rho|) \leq 2$ is sufficient for the butterfly bounds. The exponent restriction is part of the guarantee, and it is exactly the domain the fitted γ leaves in Section 7.1.4. Our implementation enforces all of these at validation time, so the SSVI contender is arbitrage-free *by the conditions it enforces*. What the data then answers is what that guarantee costs in fit, particularly where the empirical skew term structure deviates from the single power law: one global ρ forces the same skew sign and, up to φ , the same shape at every tenor. One qualification, because the results depend on it: (12) is checked at the θ -pillars, whereas Section 7.1.4 scans g on a dense (k, t) grid whose times fall between them. The two are not the same test, and on this panel they disagree in both directions: the fitted exponent leaves $(0, \frac{1}{2}]$ on 92% of captures, so the pillar conditions withhold their certificate almost everywhere, while the dense scan finds actual butterfly violations inside the quoted region on 9% of name-days.

3.3 eSSVI

eSSVI (Hendriks and Martini, 2019) keeps SSVI’s slice shape and frees its parameters expiry by expiry. Writing $\psi_t = \theta_t \varphi_t$, so that the ATM skew is exactly $\rho_t \psi_t$, each slice is

$$w_{\text{eSSVI}}(k, t) = \frac{\theta_t}{2} \left[1 + \rho_t \frac{\psi_t k}{\theta_t} + \sqrt{\left(\frac{\psi_t k}{\theta_t} + \rho_t \right)^2 + 1 - \rho_t^2} \right], \quad (13)$$

with $w(0, t) = \theta_t$ and $\partial_k w(0, t) = \rho_t \psi_t$ by construction. Setting $\rho_t \equiv \rho$ and $\psi_t = \theta_t \varphi(\theta_t)$ recovers (10) exactly, so SSVI is the constrained special case — which is what makes the pair a clean experiment, and which we exploit as a regression test on the implementation.

Three properties distinguish it from “SVI with fewer parameters”. First, taking θ_t from the quoted ATM variance rather than fitting it leaves **two** free parameters per slice — the most parsimonious of the per-expiry fits, until the greedy variant of Section 3.4 halves that again. Second, interpolation stays inside the family: the surface between pillars comes from interpolating (θ, ψ, ρ) , so every intermediate expiry is itself an eSSVI slice, whereas interpolating two SVI slices in total variance yields a smile that is not SVI. Third, the Gatheral–Jacquier butterfly bounds carry over slice-wise unchanged ($\psi(1 + |\rho|) \leq 4$ and $\psi^2/\theta(1 + |\rho|) \leq 4$), and calendar structure can be encouraged *during* calibration by fitting slices in time order under a penalty against crossing their predecessor, rather than repaired afterwards. Section 7.1.4 measures how far a penalty falls short of a guarantee, and the answer is: at the fitted pillars themselves on just over half of captures, and inside the quoted region on 15% of name-days.

Remark 1 (Scope of the no-arbitrage checks). *We enforce the per-slice butterfly bounds analytically, but check calendar arbitrage numerically — total variance non-decreasing in t at fixed k , scanned on a dense grid — rather than through the sharp pairwise analytic conditions of Hendriks and Martini (2019), as corrected by Pasquazzi (2023) (Appendix F). The numerical check is the definition, and is correct to grid resolution over the scanned region; it is not a substitute for the analytic domain if provable admissibility over the whole real line is required. Calibration approaches that work with the analytic domain directly are given by Corbetta et al. (2019) and Martini and Mingone (2022).*

3.4 The global parametrization, and the power-law backbone

The sequential calibration above enforces the no-arbitrage conditions by penalty, and Section 7.1.4 measures what a penalty leaves behind. Mingone (2022) reparameterize the N slices so that the Hendriks–Martini conditions cannot be violated by any admissible parameter vector, and calibrate all maturities in one joint least-squares problem.

For successive slices $(\rho_1, \theta_1, \psi_1)$ and $(\rho_2, \theta_2, \psi_2)$ the conditions are

$$\theta_2 > \theta_1 > 0, \quad \psi_1 p_2 < \psi_2 \leq \min\left(\frac{\psi_1}{\theta_1} \theta_2, \frac{4}{1+|\rho_2|}, \sqrt{\mathfrak{f}(\theta_2, |\rho_2|)}\right), \quad p_2 := \max\left(\frac{1+\rho_1}{1+\rho_2}, \frac{1-\rho_1}{1-\rho_2}\right), \quad (14)$$

with $\mathfrak{f}$ the butterfly bound (we use the Gatheral–Jacquier form $\mathfrak{f}_{GJ} = 4\theta/(1 + |\rho|)$; the sharper Martini–Mingone bound requires an inner minimization per evaluation). (14) is stated in the form corrected by Pasquazzi (2023), whose result and its bearing on this paper are set out in Appendix F.

Mingone’s construction takes free parameters $\rho_1, \dots, \rho_N \in (-1, 1)$, $\theta_1 > 0$, $a_2, \dots, a_N > 0$ and builds

$$\theta_i = \theta_{i-1} p_i + a_i, \quad A_{\psi_i} = \psi_{i-1} p_i, \quad C_{\psi_i} = \min\left(\frac{\psi_{i-1}}{\theta_{i-1}} \theta_i, S_i\right), \quad S_i = \min(f_i, S_{i+1}/p_{i+1}), \quad (15)$$

where f_i is slice i ’s own butterfly cap. We call the interval (A_{ψ_i}, C_{ψ_i}) the *tube*: the set of skew scales slice i may take without creating arbitrage against its predecessor. Its floor is the wing-slope requirement of (14), that the asymptotic slopes $\psi(1 \pm \rho)$ of total variance not decrease in maturity; its ceiling is the smaller of the calendar bound $\psi_{i-1}\theta_i/\theta_{i-1}$ and the butterfly cap. The name is the geometric picture: walking up the ladder of maturities, these intervals trace a corridor in (t, ψ) through which any admissible ladder must pass. The backward recursion S_i is the look-ahead: it carries the constraint of every remaining slice back to slice i , so a ψ_i chosen inside its tube can never strand a later slice. Any ψ_i strictly inside its tube then yields N slices free of butterfly and calendar-spread arbitrage, which is Mingone’s Proposition 3.1. The recursion, the argument that the tube is never empty, and our mechanical verification of the proposition are in Appendix F.

The backbone. Where Mingone (2022) places ψ_i in the tube with a free coefficient $c_i \in (0, 1)$ per slice, we tie the target to SSVI’s power-law curvature,

$$\hat{\psi}(\theta) = \eta \left(\frac{\theta}{1 + \theta} \right)^{1-\gamma}, \quad (16)$$

projected into the tube (and the projection counted: a slice whose target left the tube is reported). This restriction sacrifices generality and buys two things: the skew term structure becomes smooth in θ by construction rather than slice by slice, and γ becomes a single, interpretable dial. The value $\gamma = \frac{1}{2}$ is not arbitrary: it is the boundary of the region $\gamma \in (0, \frac{1}{2}]$ in which the Gatheral–Jacquier sufficient conditions certify the power-law SSVI free of static arbitrage, and it is the exponent that reproduces the $T^{-1/2}$ decay of the ATM skew that equity index term structures favour.

The greedy variant. With the backbone supplying ψ , the joint solve is not the only way to the guarantee — and not the cheapest. The conditions (14) are pairwise-adjacent, and calendar ordering is transitive ($w_1 \leq w_2$ and $w_2 \leq w_3$ pointwise give $w_1 \leq w_3$). A *forward* pass that keeps each slice inside its predecessor’s tube therefore guarantees the whole ladder without Mingone’s forward-looking cap: that look-ahead exists to protect a freely chosen ψ_i from stranding a later slice, and a projected backbone target needs no such protection. The S-eSSVI arm therefore proceeds greedily, taking θ_i from the quoted ATM variance and constraining ρ_i to the sub-interval on which the tube is non-empty. That interval always contains ρ_{i-1} , so it is never empty, and its edges are located numerically (Appendix F). The backbone target is then projected into the tube and the *one* remaining parameter ρ_i fitted to the slice’s quotes, with every projection and every interval binding counted. The result is the Hendriks–Martini guarantee at per-expiry cost, at the price of a backbone-constrained skew term structure and a greedy, locally optimal ρ path.

Scope of the guarantee. Proposition 3.1 covers the N fitted pillar slices (for the greedy arm, the pairwise conditions hold along the fitted ladder, which is the same statement). Between pillars the surface interpolates (θ, ψ, ρ) linearly in t exactly as the sequential fit does, and the

interpolation is *measured*, not assumed, by the same dense grid scans applied to every arm (Section 7.1.4).

3.5 How each parameterization yields a local volatility

Because the chain differentiates each fitted surface twice in k and once in t , the *derivative structure* of each is as consequential as their fit.

Strike derivatives. All three are elementary functions of k whose first and second derivatives are closed-form, so $g(k)$ and the Dupire denominator are analytic for every contender and no numerical k -differentiation enters the comparison at all. This is a deliberate feature of the design: whatever separation we measure in Section 7.2.1 cannot be attributed to one contender being differentiated numerically while another is not. What differs is not the machinery but the *function* being differentiated — and, crucially, whether the parameters of that function are free to move independently from one expiry to the next.

Time derivatives. Here the split runs the other way. The per-expiry fits (SVI, eSSVI) produce independent slices, and a surface requires interpolation. Under the linear-in- t conventions of Section 4, $\partial_t w$ is piecewise constant for the benchmark’s linear-variance rule and near-constant per segment for the parameter-interpolated arms — and for every arm it *jumps* at each pillar expiry. The induced local volatility therefore steps in time at every pillar, and the *shape* of the step in k differs on each segment, being rebuilt from a different pair of fitted smiles.

SSVI does not escape the seam, but it localizes it. Differentiating (10) through the chain rule in $\varphi(\theta)$ gives

$$\partial_t w(k, t) = \dot{\theta}_t \left[\frac{w}{\theta_t} + k \frac{\varphi'(\theta_t)}{\varphi(\theta_t)} \partial_k w \right], \quad \frac{\varphi'}{\varphi} = - \left[\frac{\gamma}{\theta} + \frac{1 - \gamma}{1 + \theta} \right] \quad (17)$$

for the power law (11). With θ_t interpolated linearly between ATM pillars — the convention we use, and the usual one — $\dot{\theta}_t$ is itself piecewise constant, so $\partial_t w$ still steps at pillars. The difference is that the bracketed shape factor is continuous in both k and t : SSVI’s step is a rescaling that is uniform across the smile, whereas a per-expiry smoother’s step changes shape. The roughness measurement of Section 7.2.1 is designed to detect exactly this mechanism.

4 Calibration and the local-volatility construction

4.1 Fitting the smiles

All fits except the SVI benchmark are Levenberg–Marquardt least squares in *unconstrained transform spaces*, so every iterate is a *domain-valid* parameter set ($\theta > 0$, $|\rho| < 1$, and so on) and no active-set logic is needed. Domain validity is a statement about parameters, not about arbitrage: only in the guaranteed arms — where the transforms map into the Mingone tube — is every iterate also an *arbitrage-admissible* slice set. The sequential eSSVI’s penalties shape the objective away from violations but guarantee nothing, at any iterate or at the optimum; how much leaks through is precisely what Section 7.1.4 measures. Arm by arm:

- **SVI** (per expiry): the quasi-explicit calibration of Zeliade Systems (2009), not Levenberg–Marquardt. At fixed (m, σ) the total-variance residuals are *linear* in $(a, d, c) = (a, \rho b \sigma, b \sigma)$,

which solve exactly as a tiny constrained least squares whose box ($a \geq 0$, $|d| \leq c$, $c + |d| \leq 4\sigma/t$) enforces non-negative variance and Lee’s wing bound by construction; only $(m, \ln \sigma)$ are searched numerically, by a deterministic grid seed refined with Nelder–Mead. The method is start-point free.

- **eSSVI** (per expiry, in time order): residuals in implied volatility; θ_t fixed to the quoted ATM total variance, leaving (ψ_t, ρ_t) free, with a butterfly penalty (weight 10) on the slice-wise bounds and a calendar penalty (weight 1 000) against crossing the previously fitted slice.
- **SSVI** (global): residuals in total variance over all quotes; η in log space, ρ through tanh, γ through a logistic map onto $(0, 1]$; θ -pillars fixed to quoted ATM total variances (not fitted), isolating the three shape parameters.
- **G-eSSVI, both arms** (global): residuals in implied volatility on the shared pillars; the Mingone parameters in unconstrained transforms (ρ_i through tanh, θ_1 and the a_i in log space anchored at the observed ATM term structure, η in log space, and — in the γ -free arm — γ through a logistic map onto $(0, 1)$), so every Levenberg–Marquardt iterate maps to an arbitrage-free slice set by construction. The standard start takes $\rho_i = 0$, the a_i from the observed ATM variance increments, and η at mid-tube.
- **S-eSSVI** (greedy): a penalty-free per-slice pre-pass sets η by closed-form regression of the free ψ_i on the backbone; the guaranteed pass then fits one bounded ρ_i per slice (Levenberg–Marquardt on the slice’s vol residuals, ρ mapped by tanh into the tube-preserving interval), with ψ_i projected onto the backbone inside the tube.

Residual spaces. The eSSVI family fits implied-volatility residuals; SVI and SSVI fit total-variance residuals. SVI has no choice, since its quasi-explicit inner solve is linear only in total variance. The eSSVI arms fit the unit they are judged in, which matters for the joint fits: because $dw = 2\sigma t d\sigma$, a total-variance objective weights each expiry’s vol errors by $(2\sigma t)^2$, so on a span from one day to 1.5 years the LEAPs would dominate and the short end go unfitted. SSVI inherits the total-variance objective from the shared SVI machinery and, being a joint fit, carries exactly that weighting, so part of its short-end fit gap is objective weighting on top of three-parameter stiffness. Reported fit errors are implied-volatility RMSE for every arm regardless, so the comparison metric is uniform even where the objectives are not.

How cost is measured. Every fit is independent: each capture is calibrated from the standard starting point above, with no information carried from any other capture. Per-capture wall clocks and iteration counts land in `timing.csv`.

One solver. Every iterative fit runs through one shared Levenberg–Marquardt implementation: pure Rust, part of the released library, single-threaded within a fit, with parameter Jacobians by forward finite differences, so one iteration of the joint fit costs its dimension-plus-one full residual evaluations. The wall clocks of Section 7.1.5 therefore measure this implementation rather than solver-independent constants, and analytic Jacobians or parallel residual evaluation could compress the joint arms’ absolute cost severalfold. What is solver-independent is the structure the timings expose, N one- and two-parameter problems against one $2N+1$ -parameter problem, so the cross-arm ratios rather than the absolute milliseconds are the claim.

Expiries with fewer than five usable quotes are skipped and counted; because the filter is applied once, before any arm sees the chain, the six arms always fit *the same* pillar set, and every count in Section 7 is therefore over identical strata. Non-converged fits are retained and flagged rather than discarded, since a practitioner’s pipeline must emit something every day.

Time interpolation. The per-expiry arms fit slices independently; a surface requires interpolation, and each arm interpolates its *natural coordinates* linearly in t . The SVI benchmark, which has no cross-expiry structure to preserve, interpolates total variance itself at fixed log-moneyness (with the forward variance floored at zero) — automatically calendar-monotone between slices, but out of family. The eSSVI arms interpolate the slice parameters (θ, ψ, ρ) and evaluate the slice formula, so every intermediate expiry is itself an eSSVI smile; because w is nonlinear in the parameters, monotonicity between two non-crossing slices is *not* automatic, which is exactly the between-pillar exposure Section 7.1.4 measures. SSVI interpolates θ_t with its global shape parameters fixed. All arms share the linear-in- t convention, proportional accrual below the first pillar, and — since every arm fits the same pillar set — a seam in $\partial_t w$ at the *same* expiries.

4.2 From fitted surface to local volatility

The local volatility is built by one shared implementation (`smoothed_surface::dupire`); the only thing that varies by model is the analytic `variance_derivatives` of the fitted surface. Guards are identical across contenders and instrumented rather than silent: the forward variance is floored, σ_{loc} is clamped to $[0.01, 3.0]$, and the fraction of grid points at which either fires is recorded per name-day. The published grid is 81×41 over $k \in [-0.4, 0.4]$ and the fitted pillar span; the fraction of it covered by live quotes is recorded so that every violation statistic can be reported twice, on the full grid and inside the quoted region. The grid is deliberately narrower than the moneyness filter ($K/F \in [0.6, 1.6]$, i.e. $k \in [-0.51, 0.47]$), so that it is bracketed by the region where quotes can exist rather than reaching the filter boundary, where only the widest-spread quotes survive. The cost of the symmetric window is a sliver of quoted put wing — $k \in [-0.51, -0.40]$, where the filter binds at longer tenors (Table 10) — that the scans do not cover; the fits use the full quoted span regardless.

This shared-machinery requirement is what makes the comparison attributable. Because every arm admits closed-form strike derivatives, none of them is handicapped by numerical differentiation, and the clamp, the guard, the grid, the interpolation rule, the finite-difference solver and the Monte Carlo draws are common.

4.3 Drift consistency

The library’s Dupire construction is written in forward log-moneyness $k = \ln(K/F(t))$. The induced local volatility is therefore consistent with (7) and with no other drift. Every simulation in this paper — the martingale test and the barrier prices alike — is driven by $d \ln F(t)/dt$ recovered from the chain’s own parity forwards, while discounting still comes from the bootstrapped Treasury curve.

One design consequence is worth stating for any replication. Simulating at the risk-free rate with zero dividends — the default in many local-volatility harnesses — aims the simulation at Se^{rT} rather than at the chain’s parity forward F_T , and on an ETF panel those two targets differ by amounts far outside Monte Carlo error on 32 768 antithetic paths. A martingale test run against any target other than the forward the surface was built in therefore measures the

harness, not the surface, and rejects surfaces that are in fact admissible. All martingale and barrier numbers in this paper drift at the parity forward.

5 Data and pipeline

5.1 Universe, collection, and the size of the panel

The panel is CBOE delayed option chains for eight liquid US names (the four most liquid index ETFs, SPY, QQQ, IWM and DIA, and four liquid single names, META, AMD, TSLA and GOOGL), collected every 15 minutes during regular trading hours over the ten sessions from 2026-08-14 to 2026-08-27, two calendar weeks. Index ETF chains are where smile parameterizations are most heavily deployed: hundreds to thousands of live quotes per capture, dense weekly and daily expiries at the short end, and listed maturities past a year. The single names extend the panel beyond the index setting: their smiles are two to three times steeper at the same tenor, their carry contains real dividends and borrow, and their early-exercise premia are large enough that the de-Americanization step below is quantitatively material. They test whether conclusions calibrated on index surfaces carry over to single-name chains. Expiries from one day to 1.5 years enter the fits, so the panel spans the ultra-short smiles and the LEAP wings in one surface.

Two slices are used, and they play different roles.

- The **daily slice** takes one capture per name-session (the capture nearest 15:30 ET) and carries the headline cross-sectional results.
- The **intraday slice** keeps every quarter-hour capture. It carries the stability instrument, and it is the robustness check on everything else: an order of magnitude more captures through the identical pipeline. More rows is not proportionally more information, because the feed is delayed and refreshes unevenly, so consecutive captures can share a quote timestamp; wherever a measurement needs a genuine refresh we select on the timestamp explicitly and report the surviving count.

Two weeks constitutes a pilot, and is treated as one. Ten sessions give nine day-over-day transitions per name. That is enough to measure a cross-section (how six calibrations of the same chain disagree today), and it is *not* enough to characterize day-over-day distributions, regime dependence, or a smoother’s stability apart from these particular weeks’ realized volatility. Day-over-day quantities are therefore stated as medians with interquartile ranges and explicit transition counts, never with a standard error that presumes independence across one name’s nine transitions, and every stability claim is made on the intraday slice, where each name contributes up to 27 snapshots per session. The design is built so a longer panel drops in without restructuring: every table and figure is generated from the same CSV schema by scripts that take the slice as an argument.

5.2 From quotes to fitting inputs

Each chain passes through identical preparation for all six arms, with every parameter fixed in the run manifest before the models are run:

1. **Discounting:** zero curve bootstrapped from US Treasury par yields for the date and rolled to the chain’s as-of date.
2. **De-Americanization:** every option in the panel is American-style, so put–call parity — which the forwards below rely on — holds only as an inequality in the raw mids. Before any implied vol is solved, each quote’s mid is lowered by its measured early-exercise premium: the parity forward is implied from the current mids, the carry $\hat{q} = r - \ln(\hat{F}/S)/t$ is read off it (measured, never assumed — it absorbs dividends, borrow and basis in one number), each quote is priced both American (CRR binomial, 201 steps) and European (Black–Scholes) at its own implied vol under $(r, \hat{q})$, and the difference, floored at zero, is subtracted from the original mid. Two rounds of this fixed-point iteration suffice; the corrected pairs then satisfy parity as an identity. The premium is a difference of two prices with matching discretization error, so it converges far faster than either price. Per-expiry outcomes (forward shifts, implied carries, premium sizes) are released in `dea.csv`.
3. **Quote filters:** quoted and uncrossed; relative spread $\leq 25\%$; $K/F \in [0.6, 1.6]$; expiries between 1 day and 1.5 years; OTM side only; at least 5 surviving quotes per expiry; implied vol inside $[0.5\%, 500\%]$.
4. **Forwards:** put–call parity on the de-Americanized mids, using the five most ATM strike pairs. The parity forward is what defines k , and — by Section 4.3 — also what defines the simulation drift.
5. **Implied vols:** safeguarded Newton with arbitrage-bound rejection; quotes violating static bounds are dropped and counted.
6. **Pre-repair:** a minimal-change static-arbitrage repair (per-expiry convex-hull projection of call prices plus a forward total-variance monotonicity sweep), applied *once*, so all three parameterizations fit the same arbitrage-clean pillar set and measured fitted-surface arbitrage is attributable to the parameterization rather than inherited from the input.

The repair statistics are themselves informative: every surviving capture arrives with butterfly violations in the raw quotes, and all but a handful with calendar violations as well (every daily capture, and 99.3% of intraday captures; `panel.csv`, columns `raw_*`). Retail-accessible delayed chains are far from arbitrage-clean, and any comparison that skips the repair step is measuring the feed, not the model.

The effect of de-Americanization is likewise measurable. The median absolute parity-forward shift it produces grows from 0.6 bp of forward inside one month to 23–24 bp at six-to-twelve months and 49 bp (index) to 62 bp (single names) beyond a year. These shifts are material relative to fit differences quoted in single basis points, and they constitute a bias rather than noise, because the American put premium always shifts the raw parity forward in the same direction. The implied carries the procedure measures are economically sensible name by name (dividend-paying DIA and IWM at +0.8 to +1.0%, the low-dividend growth names slightly negative), which serves as an internal check that the fixed point converged to the market’s carry rather than to an artefact of the trees.

5.3 Sample attrition

The cleaning cascade is input hygiene rather than a result, so its full accounting lives in Appendix B. There, Table 9 follows the raw chain rule by rule: roughly 21% of quotes reach the

fits, half of the loss being the out-of-the-money selection by design. Table 10 shows what the fits received per name and per tenor bucket: expiries and pillar counts, quotes per expiry, the quoted log-moneyness spans that anchor the surface, and the repair activity that produced the shared pillars.

One fact from that accounting must stay in the main text because the comparison rests on it. Every capture that fails does so at the *shared* cleaning stage, upstream of all six arms, so a failure removes the capture from every arm identically; no capture in either slice failed inside a fit. The comparison is therefore balanced by construction: identical name-days, identical expiry slices, identical strata for every arm.

6 The measurement chain

The evaluation follows the pipeline’s own three stages: calibrate a surface, induce a local volatility from it, and price with the local volatility. Every measurement belongs to exactly one stage, as a property of the fitted implied surface, of the induced σ_{loc} , or of the price, and each is defined where it is first reported.

One protocol is stated here rather than at the point of use, because it governs a comparison rather than a single number. In the leave-one-expiry-out test the refit sees only the remaining pillars: the held-out expiry’s parity forward is never handed to the fit, and strike conversion uses each refitted surface’s own interpolation over the pillars it retained, which is identical code for every arm. The first and last pillars are never held out, since a surface can only extrapolate to them.

Two further measurements are instruments rather than links in the chain. *Parameter stability*, taken intraday at the pillar nearest one month, sets the recalibration-noise floor against which the barrier dispersion is judged: a gap between calibrations is meaningful only if it exceeds what refitting one calibration on a barely changed chain produces. *The intraday–daily comparison* shows that the 15:30 daily cut does not drive the numbers.

Not run. The design included an unsmoothed baseline: the same chain through numerical differentiation of interpolated pillars. We did not run this baseline, so the study compares parameterizations rather than quantifying the value of smoothing itself.

7 Results

All reported figures are computed from the released CSVs by the table and figure scripts. Tables report medians with either interquartile ranges or 90th percentiles, as indicated. The daily panel contains 80 captures (8 names $\times$ 10 sessions), all surviving cleaning; the intraday panel contains 1,941 captures, of which 1,882 survive, with all exclusions occurring at 09:45 because relative spreads exceed the common cleaning cap. A median index capture has 30 expiry slices and 2,500 cleaned quotes, while a single-name capture has 21 slices and 1,100 quotes.

Raw quotes are materially non-arbitrage-free: every daily capture has butterfly and calendar violations, with a median of 266 violating triples and a median largest repair of 10.5 volatility points. All six calibration arms therefore operate on the repaired pillars rather than the raw quotes.

Table 2: **Fit quality across the six arms.** Per-slice root-mean-square residual against the shared repaired pillars, in implied-volatility basis points; medians and 90th percentiles, never means, because the per-expiry arms carry heavy tails at the shortest tenors. Bucket n is the number of expiry slices and is identical across arms, since every arm fits the same pillars; the price of each restriction is read off the ladder in Section 7.1.1.

Stratum	Slices	S-eSSVI ($\gamma=\frac{1}{2}$)		G-eSSVI ($\gamma=\frac{1}{2}$)		G-eSSVI (γ free)		eSSVI (seq.)		SSVI		SVI (benchmark)	
		Med.	p90	Med.	p90	Med.	p90	Med.	p90	Med.	p90	Med.	p90
<i>By tenor bucket (daily)</i>													
≤1W	224	135.4	432	101.8	266	113.9	257	83.6	257	445.3	736	28.2	144
1W–1M	510	145.8	361	120.6	228	122.5	231	122.3	256	283.4	468	38.2	146
1–6M	645	82.7	178	76.3	159	70.7	142	69.2	149	104.2	190	24.7	66
>6M	526	23.4	53	40.4	80	25.0	54	16.8	48	23.5	60	8.5	30
<i>By ticker (daily)</i>													
SPY	303	121.0	218	104.1	174	112.9	174	103.4	181	155.6	556	40.0	72
QQQ	296	120.7	193	103.0	167	106.0	164	96.7	170	147.1	459	24.5	60
IWM	302	70.8	129	61.6	111	61.9	111	62.7	116	138.1	388	16.9	32
DIA	182	57.7	115	43.7	80	41.8	79	35.2	62	127.4	508	8.6	43
META	204	39.7	143	46.8	115	40.8	118	34.3	119	58.4	168	21.5	55
AMD	210	103.8	328	72.5	180	63.3	185	57.6	201	126.8	364	16.8	49
TSLA	204	152.5	585	168.4	324	122.1	329	128.0	357	155.5	429	81.8	228
GOOGL	204	53.0	164	46.6	114	48.6	113	41.6	102	60.9	151	23.3	62
All daily slices	1 905	77.6	230	73.0	172	66.4	169	59.6	174	112.2	424	22.9	70
All intraday slices	44 409	74.0	214	72.1	170	64.6	164	57.2	169	108.0	433	23.0	71
<i>Solver outcome (% of fitted slices; daily / intraday)</i>													
Converged		99.3	99.0	73.9	76.5	81.8	76.4	96.8	96.6	100.0	99.9	100.0	100.0
Passes own conditions		100.0	100.0	100.0	100.0	100.0	100.0	74.3	76.8	49.6	49.6	100.0	100.0

“Converged” is each solver’s own termination flag, stopped on tolerance rather than on the iteration cap (for the global arms the single Levenberg–Marquardt flag of the joint fit, repeated on every slice row; for SVI, the outer simplex). Joint-fit convergence flags are not used as a quality criterion: alternative starting points produce materially different parameter vectors, a median $\|x\|_\infty$ distance of 100–139 in transformed units, at RMSE identical to a hundredth of a basis point. “Passes own conditions” applies each model’s own parameter-level sufficient no-arbitrage conditions to its own fitted parameters. It is a certificate the model issues about itself, not a measured violation count (Table 5 measures the surfaces), and the conditions themselves differ by model, so the column is not a like-for-like comparison across arms: the guaranteed arms pass by construction, the penalized sequential fit’s parameters leak, and SSVI fails because its fitted exponent leaves $(0, \frac{1}{2}]$. The power-law ψ target left the arbitrage-free tube and was projected onto it on a median 19.7% (S-eSSVI ($\gamma=\frac{1}{2}$)) / 3.4% (G-eSSVI ($\gamma=\frac{1}{2}$)) / 4.1% (G-eSSVI (γ free)) of slices per capture (daily).

7.1 The fitted surface and its cost

7.1.1 Fit quality

Table 2 quantifies the cost of each design choice. Raw SVI fits tightest at 22.9 bp per slice, followed by sequential eSSVI at 59.6 bp. The guaranteed construction adds 6.8 bp with a free exponent and another 6.6 bp when $\gamma = \frac{1}{2}$ is pinned. The greedy variant adds 4.6 bp, reaching 77.6 bp, as its slice-local ρ path triggers projection more often than the joint fit (19.7% against 3.4% of slices). SSVI, with three shape parameters for the full surface, is loosest at 112.2 bp. All guaranteed variants remain within roughly 18 bp of sequential eSSVI, an increment smaller than the volatility target implied by the quotes’ own bid–ask spreads; the greedy variant exceeds its ~ 10 bp design target.

The structured methods fit single names 20–30 bp tighter than the index ETFs, whereas raw SVI shows little difference (22.8 against 23.4 bp). Single-name captures contain fewer expiries

Table 3: **Out of sample: the leave-one-expiry-out test (daily slice).** Each interior pillar is held out in turn, every arm refitted on the rest, and the refitted surface evaluated at the held-out slice's strikes. In-sample and LOO RMSE are measured against the same repaired pillar vols, in implied-vol basis points, so the ratio measures overfitting directly. Reading in Section 7.1.2.

Model	n	In-sample		LOO RMSE (bp)		LOO/IS	ATM bias	LOO median by tenor (bp)		
		med. bp	med.	[IQR]	med.			$\leq 1M$	1–6M	$> 6M$
S-eSSVI ($\gamma = \frac{1}{2}$)	1 745	79.9	101.8	[49.8, 176.0]	1.12	3.6	172.4	107.4		35.8
G-eSSVI ($\gamma = \frac{1}{2}$)	1 745	74.8	97.8	[60.2, 157.1]	1.13	-47.4	150.2	104.4		53.0
G-eSSVI (γ free)	1 745	68.0	95.7	[51.3, 149.8]	1.18	-38.3	150.4	97.8		38.6
eSSVI (seq.)	1 745	62.5	97.9	[45.8, 159.1]	1.24	3.5	159.2	101.6		31.2
SSVI	1 745	113.0	134.5	[59.5, 273.7]	1.06	3.3	331.0	121.7		33.2
SVI (benchmark)	1 745	24.0	49.4	[23.4, 94.9]	1.84	5.7	85.1	50.7		19.8

One row per (capture, arm, held-out interior expiry); first and last pillars are never held out. *In-sample* is the full fit's RMSE at the same held-out expiries, so the LOO/IS ratio compares like with like (medians of per-row ratios). *ATM bias* is the signed error, model minus market, at the quote nearest the forward — negative means the interpolated smile sits below the market at the money. By family, the pooled LOO medians are 101.3 / 89.8 / 91.6 / 94.4 / 150.2 / 43.1 bp on the index ETFs and 106.5 / 119.6 / 103.3 / 104.6 / 114.5 / 79.2 bp on the single names, in display order. No holdout failed to refit on any arm. Dense short-end listings put the two neighbours of a held-out weekly a few days away, so the $\leq 1M$ column mostly tests local smoothness; the sparse buckets are where structure and flexibility separate.

(~ 21 against ~ 30), giving the global backbone fewer slices to reconcile. The result suggests that the fit cost of structure increases with the length of the term structure it must accommodate.

7.1.2 Out-of-sample interpolation

Table 3 reports the leave-one-expiry-out test of Section 6, over the 1 745 interior pillars of the daily slice. Four readings follow.

Overfitting concentrates where flexibility is. The benchmark degrades most: its median LOO error is 1.84 times its in-sample error, so half its in-sample advantage does not transfer to a smile it never saw. Every structured arm generalizes at close to its in-sample error (ratios 1.06–1.24), with the stiffest surface, SSVI, lowest.

Structure nevertheless does not achieve the lowest absolute error. SVI remains the most accurate interpolator in levels, because the structured arms' out-of-sample error is dominated by the same backbone misfit they carry in-sample rather than by interpolation noise. The informative margin is the long end: past six months, where pillars are sparse and interpolation matters most, the gaps compress from a factor of two to a few basis points and the ranking inside the family changes.

The pin's one measurable cost appears here. At the long end the pinned joint arm interpolates at 53.0 bp against the free arm's 38.6, the largest pin-versus-free gap anywhere in the paper. An exponent held away from its fitted value costs most where the backbone must carry the surface across a sparse gap unassisted, which is the out-of-sample counterpart of the misspecification the fit ladder prices at 6.6 bp.

ATM anchoring transfers out of sample; joint re-levelling does not. The signed ATM column splits the family in two. The arms that read θ off the quoted ATM variance, and the benchmark, interpolate the held-out ATM level nearly without bias; the two joint arms sit roughly 40 bp below the market. Their re-levelled θ ladder smooths through the locally elevated ATM levels

Table 4: **The fitted exponent, name by name.** Median [IQR] of the fitted power-law exponent per capture on the intraday slice, and the share of captures fitting above $\frac{1}{2}$ — for G-eSSVI, where the $\gamma = \frac{1}{2}$ pin binds; for SSVI, a guarantee diagnostic (Sections 7.1.3 and 7.1.4).

Ticker	G-eSSVI (γ free)			SSVI		
	Med.	IQR	$> \frac{1}{2}$ (%)	Med.	IQR	$> \frac{1}{2}$ (%)
SPY	0.637	[0.633, 0.639]	100	0.595	[0.592, 0.596]	100
QQQ	0.509	[0.488, 0.523]	64	0.590	[0.586, 0.594]	100
IWM	0.509	[0.501, 0.524]	79	0.597	[0.591, 0.608]	100
DIA	0.464	[0.442, 0.483]	3	0.852	[0.812, 0.868]	100
META	0.583	[0.563, 0.602]	100	0.512	[0.491, 0.530]	65
AMD	0.551	[0.531, 0.577]	95	0.551	[0.515, 0.589]	83
TSLA	0.594	[0.557, 0.624]	100	0.618	[0.599, 0.634]	100
GOOGL	0.562	[0.552, 0.571]	100	0.556	[0.533, 0.573]	92
All	0.549	[0.507, 0.593]	80	0.592	[0.556, 0.611]	92

present in the quotes, and what it smooths away in-sample it cannot recover between pillars. A desk pricing an unlisted expiry off the joint surface inherits that bias systematically, and no in-sample statistic detects it.

7.1.3 The fitted exponent

Figure 3 (third row) and Table 4 show why pinning $\gamma = \frac{1}{2}$ is inexpensive. With γ free the median is 0.549 (5th–95th percentile 0.453–0.639), close to the pinned value. The cross-name variation is more informative: DIA sits below the pin (0.46), IWM and QQQ near it (0.51), the single names range from 0.55 (AMD) to 0.59 (TSLA), and SPY is highest at 0.64. The pin is therefore broadly reasonable, but it understates the slower skew decay of every single name in this panel, and most of its fit cost is concentrated in SPY. That is a property of these four names rather than of single names generally: Appendix D fits the exponent on six further names and finds two of them below $\frac{1}{2}$.

Projection onto the arbitrage-free tube is modest in the joint fits, affecting 3.4% of slices in the pinned arm and 4.1% in the free arm. It is substantially more frequent in the greedy arm (19.7%), where fixing θ to the observed ATM term structure and choosing ρ slice by slice leaves a narrower tube for the backbone.

7.1.4 Static arbitrage

Table 5 counts fitted name-days with at least one butterfly violation or calendar crossing, on the full 81×41 grid ($k \in [-0.4, 0.4]$ across the fitted pillar span) and separately inside the quoted region. It reports counts and rates only: the severity measure $\min g$ is dominated by degenerate benchmark slices, and how far such a fit travels past the boundary is not a meaningful magnitude.

The construction holds exactly, greedy or joint. At the fitted pillar slices all three guaranteed arms show a measured maximum calendar crossing of 0.0, zero to machine precision, on every capture of both slices, and zero butterfly violations inside the quoted region throughout. This is Proposition 3.1 of Mingone (2022) confirmed on market data for the joint arms, and the pairwise tube argument of Section 3.4 for S-eSSVI: the guarantee survives the greedy shortcut exactly.

Table 5: **Static arbitrage of the fitted surfaces.** Share of fitted name-days with at least one butterfly violation or calendar crossing, on the full grid and restricted to the quoted region; counts and rates only. The last pair of columns is different in kind: it applies each model’s own sufficient no-arbitrage conditions to its own fitted parameters, so it is a certificate the model issues about itself rather than a measured violation count, and the conditions differ by model. Scope of the scans and how to read them: Section 7.1.4.

Model	Days	Butterfly full grid		Calendar full grid		Either, inside quotes		Passes own conditions	
		<i>n</i>	%	<i>n</i>	%	<i>n</i>	%	<i>n</i>	%
<i>Daily slice</i>									
S-eSSVI ($\gamma=\frac{1}{2}$)	80	0	0.0	11	13.8	10	12.5	80	100.0
G-eSSVI ($\gamma=\frac{1}{2}$)	80	0	0.0	8	10.0	1	1.2	80	100.0
G-eSSVI (γ free)	80	0	0.0	10	12.5	1	1.2	80	100.0
eSSVI (seq.)	80	0	0.0	18	22.5	12	15.0	62	77.5
SSVI	80	8	10.0	0	0.0	7	8.8	44	55.0
SVI (benchmark)	80	8	10.0	45	56.2	16	20.0	80	100.0
<i>Intraday slice</i>									
S-eSSVI ($\gamma=\frac{1}{2}$)	1 882	0	0.0	269	14.3	240	12.8	1 882	100.0
G-eSSVI ($\gamma=\frac{1}{2}$)	1 882	0	0.0	201	10.7	101	5.4	1 882	100.0
G-eSSVI (γ free)	1 882	0	0.0	230	12.2	119	6.3	1 882	100.0
eSSVI (seq.)	1 882	0	0.0	418	22.2	279	14.8	1 498	79.6
SSVI	1 882	206	10.9	0	0.0	185	9.8	1 037	55.1
SVI (benchmark)	1 882	181	9.6	1 045	55.5	406	21.6	1 882	100.0

The sequential penalty, by contrast, leaves residual pillar crossings on 43 of 80 daily captures (up to 4×10^{-5} in total variance) and at least one calendar crossing inside the quoted region on 15% of name-days.

What the guarantee does not cover is measured. On the dense (k, t) grid the joint arms show calendar crossings on 10–12% of name-days, 1% inside the quoted region, and the greedy arm on 14%, 12% inside quotes and close to the penalty arm’s 15%. Every one of these lives *between* pillars, where the surface interpolates (θ, ψ, ρ) linearly in t and two arbitrage-free slices do not guarantee an arbitrage-free interpolant. The greedy arm’s higher interior rate follows from its slice-local ρ path, which moves more between adjacent expiries than the jointly negotiated one (Figure 3, bottom row). The pillar guarantee is identical across the three arms; the between-pillar interpolation is measurably better under the joint solve.

SSVI’s conditional guarantee does not bind. The fitted SSVI exponent sits above $\frac{1}{2}$ on 92% of captures (Table 4), outside the domain $\gamma \in (0, \frac{1}{2}]$ in which the Gatheral–Jacquier conditions certify the surface. The consequence is visible three ways: SSVI fails its own conditions on those captures, shows butterfly violations *inside the quoted region* on 9% of name-days, more than any other arm including the benchmark, and does so while fitting worst.

The benchmark behaves as expected. SVI shows calendar crossings on 56% of name-days on the full grid (19% inside quotes) and butterfly violations on 10%, and it is the only arm whose local volatility pins the upper Dupire clamp on any capture.

Table 6: **What each calibration costs.** Wall-clock milliseconds per capture (median [IQR]) for one independent fit from the standard starting point, single-threaded within a fit, on the study machine, split by family because the joint problem scales with the expiry count: a median 60-parameter problem on the ~ 30 -expiry index chains against 43 on the ~ 21 -expiry single names. The sequential eSSVI and SVI solve one small problem per expiry, and SSVI one three-parameter problem (ρ, η, γ ; the θ pillars are read off the quotes, not fitted).

Model	Index ETFs				Single names			
	Dim	ms	IQR	Iter.	Dim	ms	IQR	Iter.
<i>Daily slice</i>								
S-eSSVI ($\gamma=\frac{1}{2}$)	—	4	[2, 6]	140	—	2	[2, 2]	80
G-eSSVI ($\gamma=\frac{1}{2}$)	60	326	[200, 763]	31	43	23	[18, 25]	7
G-eSSVI (γ free)	61	319	[187, 450]	28	44	26	[22, 31]	8
eSSVI (seq.)	—	5	[4, 7]	—	—	2	[2, 2]	—
SSVI	—	14	[8, 20]	—	—	6	[5, 8]	—
SVI (benchmark)	—	1 164	[698, 1 345]	—	—	547	[467, 631]	—
<i>Intraday slice</i>								
S-eSSVI ($\gamma=\frac{1}{2}$)	—	4	[2, 6]	134	—	2	[1, 2]	79
G-eSSVI ($\gamma=\frac{1}{2}$)	59	390	[241, 581]	31	41	22	[17, 27]	7
G-eSSVI (γ free)	60	307	[201, 504]	28	42	26	[20, 34]	8
eSSVI (seq.)	—	4	[4, 6]	—	—	2	[2, 2]	—
SSVI	—	12	[8, 20]	—	—	6	[4, 7]	—
SVI (benchmark)	—	1 084	[692, 1 217]	—	—	536	[460, 623]	—

The IQR $[a, b]$ is the interquartile range — the 25th-to-75th percentile span, holding the middle half of all captures. Medians and IQRs are used here and throughout, never means and standard deviations, because every distribution summarized is skewed (wall clocks by pillar counts and iteration caps, fit errors by short-tenor tails): a mean would be dominated by tails that say nothing about the typical capture. The IQR describes spread, not the uncertainty of the median. Dimension of the global problem: $2N+1$ parameters for N expiries with γ fixed, one more with γ free. Iterations are Levenberg–Marquardt steps. Where the pin is misspecified the optimizer pays in iterations, not per-iteration cost: on SPY — the name whose free exponent strays furthest from $\frac{1}{2}$ — the pinned arm runs a median 31 iterations and 814ms against the free arm's 22 and 392ms (daily), grinding along the misspecification floor despite one fewer parameter.

7.1.5 Calibration cost

Table 6 reports the guarantee’s cost in milliseconds, split by family because the joint problem scales with the expiry count. *S-eSSVI is the fastest calibration in the panel*: a median 4.2 ms on the index chains and 2.0 ms on the single names, ahead of the penalty sequential fit it replaces on the index half (5.3 ms) and tied on the singles, because its per-slice problem has one free parameter against the penalized fit’s two. The construction that carries the same pillar guarantee through a joint solve costs a median ~ 320 ms on a thirty-expiry index chain and ~ 25 ms on a twenty-one-expiry single name, roughly $60\times$ the sequential fits and $20\times$ SSVI. The Hendriks–Martini guarantee therefore does not require the joint solve’s orders of magnitude, only the backbone and the forward pass.

One anomaly merits discussion: the pinned arm is slower than the free one, despite one fewer parameter, exactly where the pin is misspecified. Per-iteration cost is essentially identical, so any gap is iteration count. The iteration gap concentrates on the index half (389 against 307 ms intraday) and above all on SPY, the name whose fitted exponent is $\gamma \approx 0.64$: there the pinned arm requires a median 31 iterations and 814 ms against the free arm’s 22 and 392 ms on the daily slice, because the residuals cannot fall below the misspecification floor and Levenberg–Marquardt progresses slowly along a shallow valley. On the single names, whose exponents sit closer to the pin and whose problems are half the size, the two joint arms are a statistical tie.

Two context numbers place the joint solve’s cost in perspective. First, the SVI benchmark is the *slowest* arm in the study, at 1.16 s on a median index capture: thirty per-slice quasi-explicit fits, each running a gridded two-dimensional search over $(m, \ln \sigma)$ with an exact inner solve per trial (Section 4), are also expensive. The joint eSSVI is intermediate in absolute terms, and every arm is comfortably inside an intraday refresh budget. Second, the entire six-arm daily study (11 430 slice fits, 480 local-volatility builds, 1 440 martingale checks and 20 160 Monte Carlo barrier and vanilla prices) completes in 305 seconds on one laptop.

7.2 The induced local volatility

7.2.1 Roughness and the round trip

Table 7 and Figure 1 qualify the pilot’s headline. The benchmark is the roughest surface (5.6×10^{-4} median mean $|\Delta_k^2 \sigma_{\text{loc}}|$, 1.8 to 2.7 times the structured arms), but among the structured arms the spread is narrow (2.1–3.0), SSVI sits *inside* the eSSVI family rather than below it, and the pooled ranking is led by the greedy arm. The per-name columns show why. On the index half of the panel a surface carries ~ 30 pillars and the shared linear-in- t interpolation makes $\partial_t w$ jump at every one, so the seam term, identical across arms, dominates the parameterization’s own contribution: every structured arm lies between 3.0 and 6.0 on the three liquid index names, and the name effect is as large as the arm effect. The single names contradict the intuition that steep smiles produce rough local volatilities. With ~ 21 sparser pillars every arm drops to 0.6–2.5, benchmark and backbone alike, and SVI alone stays at 2.8–4.0.

The discriminating case is the thin book. On DIA the benchmark’s roughness rises to 13.8, SSVI’s to 8.3 and the penalty eSSVI’s to 6.1, while the three guaranteed arms hold 3.5–4.4. What separates DIA from the single names is instructive: both are thin relative to SPY, but the single names’ ladders are sparse and clean where DIA’s is dense and noisy. It is quote noise in the θ increments, not pillar count, that the weaker arms amplify, and structure substitutes for data exactly there.

A second pattern runs across the index columns. The two arms that anchor θ to the *quoted*

Table 7: **The induced Dupire local volatility, name by name.** Panel A: curvature roughness of σ_{loc} per ticker. Panel B: clamp activity against the $[0.01, 3.0]$ bounds, the local-vol range, and the finite-difference round trip back to each arm's own smile. Definitions in the footnote; reading in Section 7.2.1.

<i>Panel A: roughness $\times 10^4$, median by name (daily slice)</i>									
Model	SPY	QQQ	IWM	DIA	META	AMD	TSLA	GOOGL	All [IQR]
S-eSSVI ($\gamma=\frac{1}{2}$)	4.66	3.12	3.08	4.35	1.00	0.64	1.38	1.13 2.09	[1.12, 3.23]
G-eSSVI ($\gamma=\frac{1}{2}$)	5.94	3.65	3.51	3.48	1.22	1.05	2.47	1.41 2.95	[1.36, 3.65]
G-eSSVI (γ free)	6.03	3.65	3.50	3.67	1.20	1.01	2.37	1.47 3.02	[1.41, 3.67]
eSSVI (seq.)	4.54	2.97	3.11	6.08	1.28	0.97	1.68	1.47 2.46	[1.51, 4.23]
SSVI	5.83	3.84	3.74	8.34	0.88	0.67	1.65	1.15 2.45	[1.05, 4.11]
SVI (benchmark)	6.29	5.98	6.92	13.80	3.70	3.95	2.84	3.58 5.57	[3.42, 9.37]

<i>Panel B: containment and round trip</i>									
Model	Daily slice				Intraday slice				
	Clamped	$\sigma_{\min}$	$\sigma_{\max}$	RT bp	Rough. All [IQR]	Clamped	$\sigma_{\max}$	RT bp	
S-eSSVI ($\gamma=\frac{1}{2}$)	0.00%	0.154	1.602	37.9	2.00 [1.11, 3.27]	0.00%	1.695	40.9	
G-eSSVI ($\gamma=\frac{1}{2}$)	0.00%	0.145	1.700	37.6	2.99 [1.34, 3.72]	0.00%	1.770	41.3	
G-eSSVI (γ free)	0.00%	0.141	1.774	38.1	3.07 [1.39, 3.74]	0.00%	1.848	40.9	
eSSVI (seq.)	0.00%	0.058	1.552	39.9	2.54 [1.46, 4.24]	0.00%	1.645	43.2	
SSVI	0.00%	0.158	1.472	47.3	2.24 [1.02, 4.14]	0.00%	1.528	52.1	
SVI (benchmark)	0.81%	0.010	2.278	36.9	5.66 [3.41, 8.77]	0.99%	2.497	40.6	

Roughness is $10^4 \times$ the mean absolute second difference of σ_{loc} along k at $\Delta k = 0.01$ over the shared grid — zero where the local-vol smile is locally linear, large where it kinks or oscillates; a value of 3 means neighbouring-strike deviations of about 3 vol basis points from the local line (raw CSV column `roughness_mean_abs_d2`). *Clamped grid* is the median share of grid cells pinned at the $[0.01, 3.0]$ bounds. Pooled across arms, the median clamped share is 3.22% on DIA and at most 0.00% on every other name. *Local-vol range*: $\sigma_{\min}$ and $\sigma_{\max}$ are medians of the per-day extremes, so day-level tail events do not appear in them: only the benchmark ever pins the 3.0 cap, outright, on 22 of 80 name-days (Figure 4c). *Round trip* (RT bp): Crank–Nicolson local-vol repricing of the calibrating vanillas, re-inverted to implied vol, against the arm's own fitted smile; against the market pillars instead the daily medians are 68.2 / 74.5 / 68.3 / 67.1 / 75.3 / 48.1 bp in display order. The `ur_*` usability columns are omitted: `usability_report` takes no carry argument and so assumes zero dividends, unlike every other column here.

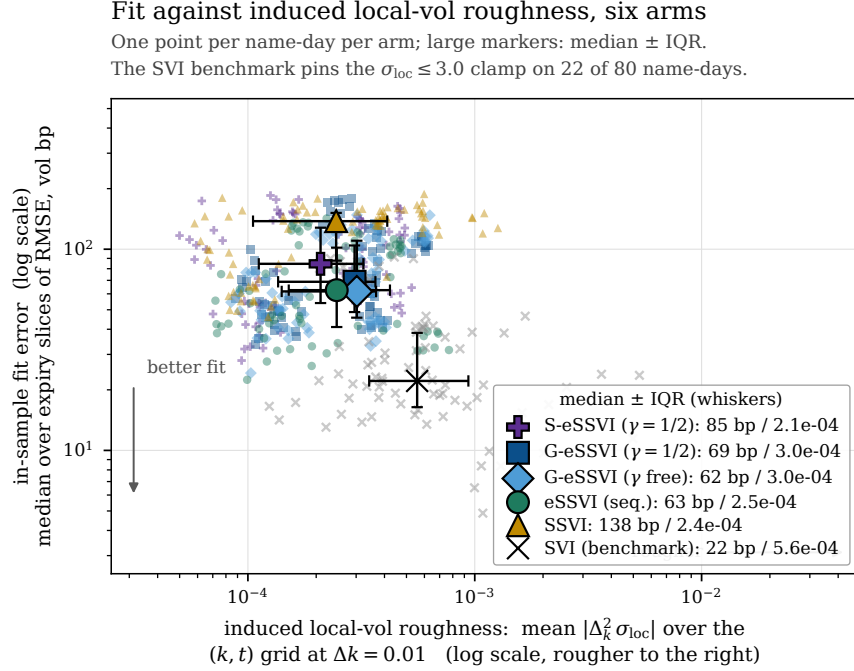


Figure 1: **Fit against induced local-volatility roughness, six arms (daily slice).** One point per name-day per arm; large markers are medians with IQR whiskers, log axes. The benchmark sits bottom-right (best fit, roughest surface) and SSVI top-centre (worst fit, and *not* the smoothest surface); the penalty and joint eSSVI arms are nearly coincident, and the greedy arm sits left of them, smoothest pooled, an advantage attributable to the noisy thin book (Table 7). On dense index term structures the roughness axis compresses, because the shared pillar-seam structure of $\partial_t w$ in the ~ 30 -pillar interpolation dominates the parameterization's own contribution, while the sparse single-name ladders pull every arm's roughness down together.

ATM variance, the sequential fit and greedy S-eSSVI, are the smoothest on the three liquid index names (3.0–4.7 against the joint arms' 3.5–6.0), and the ranking reverses on DIA, where the joint arms are smoother (3.5–3.7 against 4.4 and 6.1). The mechanism sits in the Dupire numerator: $\partial_t w$ is driven by the increments of the ATM term structure, so on a liquid name taking the already-smooth quoted θ verbatim leaves less for the seams to differentiate than a jointly re-levelled θ , while on the thin name the quoted θ is itself noisy and the joint solve's smoothing pays. The pooled medians place the greedy arm smoothest overall (2.09), though the IQRs overlap and the separation is suggestive rather than sharp. The joint arms remain indistinguishable (2.95 against 3.02 pooled, and within 0.2 on every name).

The finite-difference round trip does not discriminate. Within the eSSVI family and the benchmark it spans 36.9–39.9 bp, each inside its own IQR, and SSVI's 47.3 bp sits only a quarter above the family. The round trip tracks *fit error* rather than local-volatility quality: it flags the arm whose smile is farthest from the quotes and is blind to everything the Dupire construction does afterwards.

7.2.2 The martingale gate

All six local volatilities pass the check, identically: across 1 440 daily and 33 876 intraday arm-horizon cells the median $|\mathbb{E}[S_T]/F_T - 1|$ is 1.38 and 1.40 bp respectively, and *zero* cells exceed $|z| = 3$, fewer than normality alone would produce. The check functions purely as a gate: it validates the barrier comparison and separates nothing, as the drift-consistency convention of Section 4.3 implies it must, since every arm is simulated at the forward its own surface was built in. The per-arm detail, pooled across horizons, is Table 11 in Appendix B.

7.3 The barrier price

Table 8 and Figure 2 report the chain’s final link. Dispersion is $(\max - \min)/\text{mean}$ across the five practical arms, the SVI benchmark excluded, with a vanilla call priced on the identical paths as a control. It is read against the recalibration-noise floor measured by the stability instrument: refitting the *same* arm on the next quarter-hour snapshot moves a one-month down-and-out put by 0.66–0.94% at the median, depending on the arm.

The five-arm menu disagrees by 11.6% of premium on the one-month down-and-out put (7.7 bp of spot). The vanilla control does not behave as a control should, and the failure is informative: it disperses 3.6% across the menu, and in money terms its 9.6 bp of spot *exceeds* the knock-out’s 7.7, because SSVI’s 112 bp fit gap mis-prices the vanillas themselves. The menu spread therefore mostly measures SSVI’s distance from the quotes, not a barrier-specific divergence of wing treatments.

The informative comparisons are the pairwise ones, where fit gaps cancel.

- **The γ question:** pinning $\gamma = \frac{1}{2}$ moves the one-month down-and-out put by a median 0.91%, inside one Monte Carlo standard error and at the refit-noise floor itself. Pinning γ therefore costs 6.6 bp of in-sample fit and no pricing accuracy that could be realised in a trade.
- **The enforcement question:** replacing the sequential penalty calibration with the global construction moves the same price by a median 2.87%, roughly three times the γ effect and four times the refit noise, at two and a half Monte Carlo standard errors. Guaranteeing the no-arbitrage conditions is not a re-labelling of the same surface; it is a different surface, and the difference is priced. This contrast is less cleanly identified than the two either side of it, since the guaranteed arms also impose the power-law backbone that the penalized fit does not, so 2.87% bounds the enforcement effect from above rather than measuring it.
- **The greedy question,** which is the paper’s most tightly controlled comparison: the two arms differ in nothing but the algorithm that locates their parameters. Calibrating the guaranteed surface greedily instead of jointly moves the same price by a median 3.78%, at three and a half standard errors and four to five times the refit noise. This is the largest pairwise gap in the family, larger than the enforcement effect (3.8% against 2.9%) and larger than the greedy arm’s own gap to the penalty fit (2.6%) — and it is the gap the design pins down most precisely, where the enforcement figure it exceeds is itself an upper bound. The vanilla controls locate the mechanism: greedy and penalty agree on vanillas to 0.1%, both anchoring θ to the quoted ATM level, while greedy and joint disagree by 2.1% even on vanillas because the joint fit re-levels θ everywhere, the same re-levelling the leave-one-out test measures as a –40 bp ATM bias (Section 7.1.2). A different ρ path and a harder-projected ψ reshape the wings that the knock-out reads and the vanilla does not.

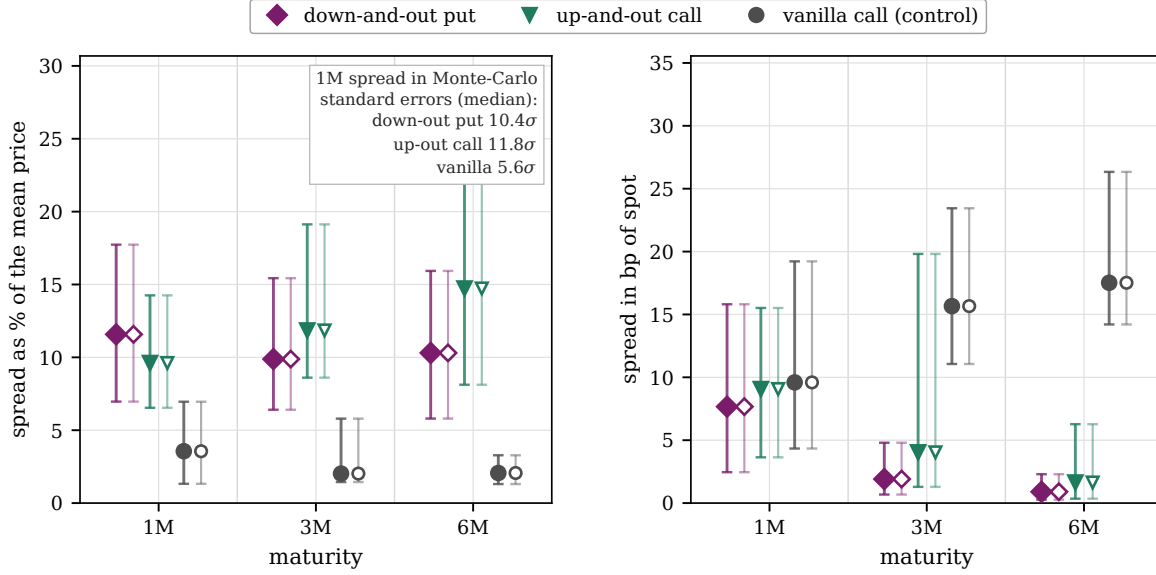
Table 8: **Barrier-price dispersion across calibrations.** Panel A: cross-arm dispersion ($\max - \min$)/mean over the five-arm practical menu (the three guaranteed eSSVI arms, sequential eSSVI, SSVI — the SVI benchmark excluded), capture by capture on one shared path set, calibrated horizons only. Panel B: the three pairwise one-variable gaps, read in Section 7.3. Medians with IQR; 40 000 antithetic paths per price, seed 20260824, identical draws across arms.

<i>Panel A: five-arm menu dispersion, calibrated horizons (daily)</i>							
Product	Mat.	n	Med. %	IQR	p90	Abs. bp	MC σ
Down-and-out put	1M	240	11.6	7.0, 17.7	41.9	7.7	10.4
	3M	240	9.9	6.4, 15.4	23.7	1.9	4.1
	6M	240	10.3	5.8, 15.9	26.3	0.9	2.4
Up-and-out call	1M	240	9.6	6.5, 14.3	20.9	9.1	11.8
	3M	240	11.8	8.6, 19.1	30.5	4.0	8.6
	6M	240	14.7	8.1, 22.7	37.0	1.6	4.1
Vanilla call	1M	80	3.6	1.3, 7.0	22.5	9.6	5.6
	3M	80	2.0	1.4, 5.8	8.6	15.7	4.3
	6M	80	2.1	1.3, 3.3	5.3	17.5	3.6
<i>Panel B: pairwise gaps, calibrated horizons (daily)</i>							
Comparison	Mat.	n	Med. %	IQR	p90	Abs. bp	MC σ
$\gamma=\frac{1}{2}$ vs γ free (global)	Down-and-out put	240	0.9	0.4, 2.6	6.3	0.5	0.8
	Up-and-out call	240	1.0	0.4, 3.1	6.2	0.7	1.2
	Vanilla call	80	0.6	0.2, 1.0	3.8	1.5	1.0
Global $\gamma=\frac{1}{2}$ vs penalty seq.	Down-and-out put	240	2.9	1.4, 5.1	7.5	1.3	2.5
	Up-and-out call	240	3.8	1.7, 6.1	8.4	2.5	4.7
	Vanilla call	80	1.8	0.7, 3.2	4.4	4.1	3.5
Greedy vs joint (both $\gamma=\frac{1}{2}$)	Down-and-out put	240	3.8	2.2, 6.5	11.3	2.1	3.6
	Up-and-out call	240	4.6	2.8, 8.1	13.3	4.0	6.6
	Vanilla call	80	2.1	0.7, 3.3	4.3	4.2	3.8
Greedy $\gamma=\frac{1}{2}$ vs penalty seq.	Down-and-out put	240	2.6	0.9, 5.8	9.5	1.2	2.1
	Up-and-out call	240	3.4	0.9, 6.7	12.1	1.7	3.2
	Vanilla call	80	0.1	0.0, 0.2	0.3	0.3	0.2

Including the SVI benchmark widens the one-month down-and-out put dispersion to 26.4% (six arms). On the intraday slice the same five-arm menu median is 11.1% ($n = 5\,643$). Panel B maturities beyond 1M follow the same ordering and are omitted for space; the Panel B “Mat.” column lists the product at 1M throughout. Knock-out rows pool three barrier levels; strike at spot. The two knock-outs reported read the two wings of the smile; the mirrored families (down-and-out call, up-and-out put) and the vanilla put behave alike and remain in the released `barriers.csv`.

Cross-arm barrier spread, in ratio and in money

Daily slice, medians with IQR whiskers. Filled: calibrated horizons. Open: all rows, incl. extrapolated.



1M, calibrated horizons, five-arm menu: the down-and-out put's spread is 11.6% of its own price against 3.6% for the vanilla, but in bp of spot the same spread is 7.7 bp against the vanilla's 9.6 bp. The multiple is large partly because the knock-out is cheap, not because more money is at stake. The SVI benchmark is excluded from the spread; including it widens every knock-out median. Medians only.

Figure 2: **Cross-arm barrier dispersion, in ratio and in money (daily slice, five-arm menu).** Median spread with IQR whiskers, by product and maturity; filled markers restrict to calibrated horizons. The vanilla control disperses materially in its own right, because SSVI's fit gap mis-prices the vanillas, so the menu spread mostly measures distance from the quotes rather than a barrier-specific divergence; the pairwise gaps in the text are the informative comparison.

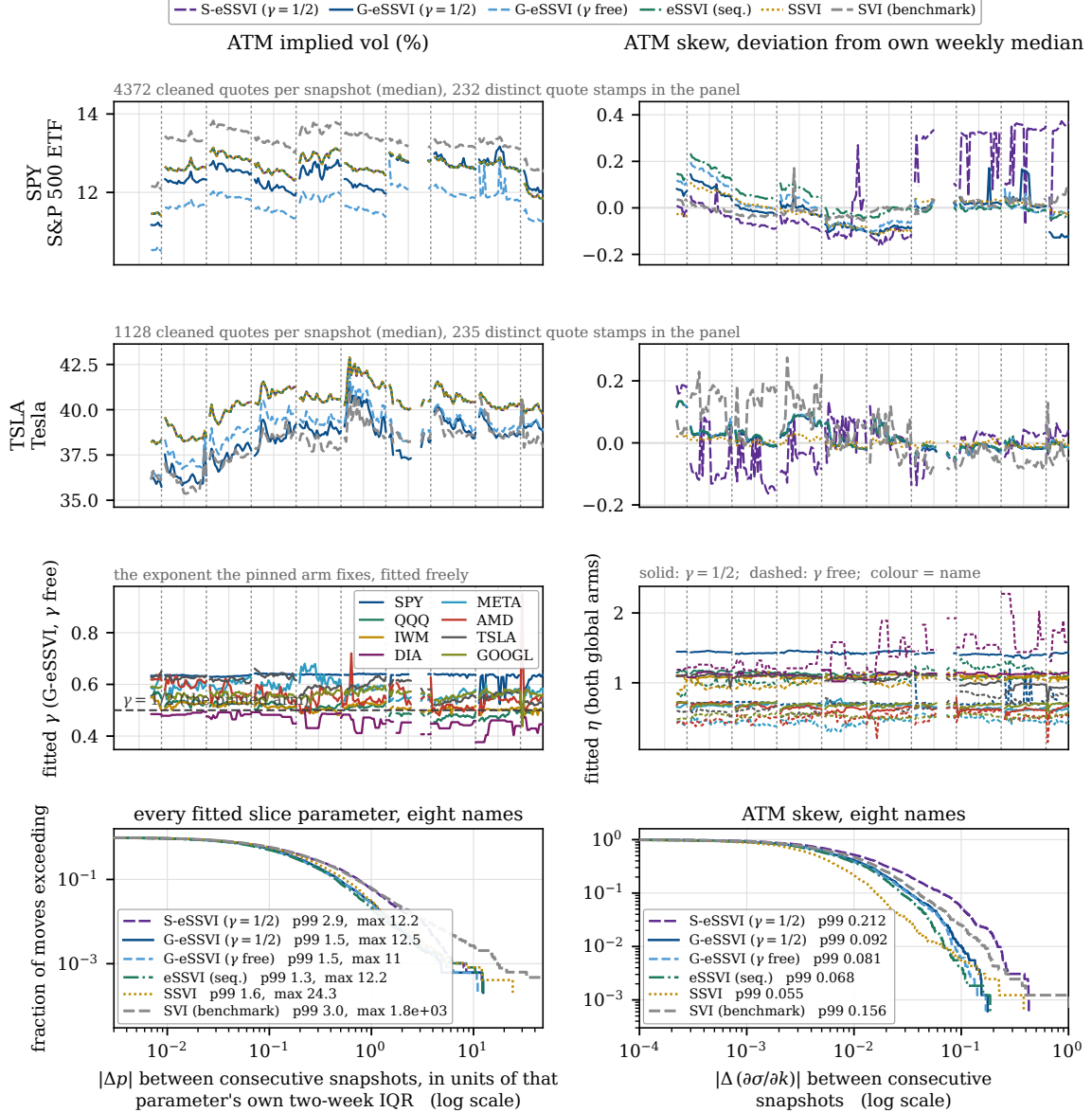
The stability instrument (Figure 3) qualifies the pilot's conclusion in both directions. On the index chains the benchmark's parameter paths are stable, its worst within-session move being 6.9 two-week IQRs, so parameter instability, the classic argument against raw SVI, largely disappears where the chains are dense. On the single names it returns: the worst move in the panel is AMD's SVI b jumping 1800 IQRs between consecutive snapshots, against 12–24 IQRs for every structured arm on the same names. Structure is what bounds the parameter tails; liquidity only conceals their absence. The one structured arm the instrument flags is the greedy one, whose snapshot-to-snapshot ATM skew moves carry a p99 of 0.21 against 0.06–0.16 elsewhere, the time-series counterpart of the slice-local ρ freedom that appears as interior calendar crossings and as the 3.8% barrier gap. Its barrier refit noise (0.94%) stays within half again the family's, so the instability resides in the parameters more than in the price.

8 Discussion

Most of the chain behaved as the design anticipated: the fit ladder is monotone in structure, the guaranteed arms are exactly clean at their pillars while the penalty calibration leaves residual violations, the γ pin is nearly immaterial at almost every link, and the round trip does not

Fitted parameters through two weeks

Intraday panel, expiry pillar nearest one month, up to 27 snapshots per session over 10 sessions. The sequential eSSVI and SSVI read the ATM anchor θ_t off the quotes, while the global arms fit it jointly; SVI's ATM level is a fitted by-product. Skew is drawn as a deviation from each arm's own panel median so the paths share one axis.



Row 3, left: the gamma-free arm's fitted exponent sits at median 0.549 (5th-95th pct 0.453-0.639) across all names and snapshots, against the 1/2 the pinned arm imposes; where the two arms differ downstream, this gap is the entire cause. Survival curves pool within-session consecutive-snapshot moves over the eight names; each parameter is scaled by its own two-week IQR before pooling.

Figure 3: **Fitted parameters through two weeks (intraday slice)**. Rows 1–2: ATM implied vol and ATM skew at the pillar nearest one month for SPY and TSLA, six arms; vertical rules are overnights. Row 3, left: the exponent the pinned arm fixes, fitted freely — DIA below $\frac{1}{2}$, IWM and QQQ just above, the four single names in a band from 0.54 to 0.60, and SPY highest near 0.64; right: the fitted backbone level η for both global arms. Row 4: survival functions of within-session snapshot-to-snapshot moves — every fitted slice parameter scaled by its own two-week IQR (left) and the ATM skew itself (right). On the index chains even the benchmark's parameter tails stay bounded (worst move 6.9 IQRs), but the single names restore the classic SVI instability: AMD's b jumps 1800 IQRs between two snapshots. The greedy arm has the heaviest skew tail in the panel (p99 move 0.21 against 0.06–0.16 for the others): its slice-local ρ re-decides each snapshot what the joint solve negotiates once.

discriminate. Five results did not, and each is reported in Section 7 where it was measured. SSVI is not the smoothest surface; the vanilla control disperses more of spot than the knock-outs it was meant to control for; the single names, expected to be the stress case, are where every arm’s local volatility is smoothest; the joint arms carry an out-of-sample at-the-money bias that no in-sample statistic detects; and the greedy variant, designed as an inexpensive substitute for the joint fit, prices the knock-out furthest from it. That last result comes from the design’s most tightly controlled contrast, which is why it carries the most weight.

8.1 Practical implications

Four implications follow, stated as the two-week panel supports them.

The guaranteed construction justifies its cost, and that cost is now a choice. It eliminates the penalty calibration’s calendar leakage exactly rather than approximately, and the difference re-prices a one-month knock-out by 2.9%. The guarantee is available through the joint solve at ~ 320 ms on an index chain, or at 2 ms through the greedy pass, which is cheaper than the penalty calibration it replaces.

Greedy and joint calibration are not interchangeable. They deliver the identical pillar guarantee, but the greedy pass costs 5 bp of fit against the joint solve, leaves penalty-level interior crossings, carries the least stable skew path in the panel, and prices the knock-out 3.8% away; against that, its quoted- θ anchor prices held-out expiries without the joint arms’ -40 bp bias. The results suggest that S-eSSVI may be suitable where speed is prioritized, while joint calibration is preferable when downstream exotic pricing is sensitive to the surface. What they do not support is treating the two as one surface at different speeds.

The exponent is a specification choice rather than a pricing one. Mingone’s guarantee is γ -independent, so the domain $(0, \frac{1}{2}]$ that disciplines SSVI’s conditional certificate adds nothing here. The pin costs 6.6 bp of fit, moves the barrier by 0.91%, and interpolates held-out long-end smiles at 53 bp against the free arm’s 39. The results favour a free exponent when fit and long-end interpolation accuracy are prioritized; pinning provides a simpler, more stable specification at modest in-sample cost. All four single names in this panel fit above $\frac{1}{2}$, so for them the pin is a mild systematic misspecification rather than a neutral default, though a wider cross-section (Appendix D) shows that is not a general single-name property.

A conditional guarantee is not a guarantee unless the calibration enforces it. On 92% of captures SSVI’s fitted γ left the sufficient-condition domain, the surface failed its own validator, and butterfly violations appeared inside the quoted region on more name-days than for any other arm.

8.2 Limitations

Five, in decreasing order of importance. (i) Two weeks, one regime, eight names — four correlated index products and four mega-cap single names: nothing here speaks to stressed markets, to small or illiquid single names, or to day-over-day distributions, and the panel is built to be extended rather than to settle those questions. (ii) The between-pillar interpolation is the binding gap in the no-arbitrage story: the guaranteed arms’ guarantee is exact at pillars, and 10–14% of their name-days still show an interpolation crossing somewhere on the dense grid. An arbitrage-free interpolant would close it. (iii) The power-law backbone is our restriction of Mingone (2022), chosen to make the γ question one-dimensional; her free- c_i parametrization might fit better and was not run. (iv) There is no unsmoothed baseline arm, so the results

compare parameterizations rather than quantifying the value of smoothing itself. (v) Two robustness variants were not run: vega-weighted residuals, and monotone C^1 time interpolation. The latter is relevant because the linear-in- t convention introduces derivative discontinuities and makes absolute roughness levels interpolation-dependent; the effect is shared across arms, so it does not explain differences between them.

8.3 Extensions

Sixty sessions would turn the day-over-day decomposition from an order of magnitude into an estimate, put error bars on the enforcement gap by regime, and test whether SPY’s $\gamma \approx 0.64$, and the dispersion of fitted exponents across names (Appendix D), is a stable feature of the term structures or an artefact of two weeks. The pipeline takes the extension unchanged: the collector keeps running, and every table and figure regenerates from the same schema.

9 Conclusion

We followed six calibrations of two weeks of chains on eight names, every quote de-Americanized, from the quotes to a barrier price, with out-of-sample interpolation and calibration cost measured on the same footing as the fit.

The global eSSVI of [Mingone \(2022\)](#), implemented here with a power-law backbone, delivers what its proposition promises: zero butterfly violations and zero calendar crossings at every fitted pillar slice, on every capture, where the sequential penalty calibration it replaces leaves residual crossings on just over half of them. That guarantee costs 7 to 13 basis points of fit and a joint solve, and it re-prices a one-month knock-out by 2.9%, so enforcement is a pricing decision rather than an implementation detail.

The guarantee does not require the joint solve. A greedy forward pass with one fitted parameter per slice (Section 3.4) delivers zero pillar violations at 2 milliseconds, the fastest calibration in the panel and cheaper than the penalty fit it replaces. It is not, however, the joint surface at lower cost: it gives up 18 basis points of fit, keeps penalty-level crossings between pillars, and prices the knock-out 3.8% away from the joint solve at the same exponent. Those two arms share everything but the algorithm that locates their parameters, so the gap isolates calibration mode: on this panel, *how* the guaranteed surface is calibrated moves the exotic price more than whether the guarantee is enforced at all.

Fit alone does not identify the better surface. Held out of sample the benchmark loses nearly half its in-sample advantage, the structured arms generalize at theirs, and the jointly calibrated arms reveal a -40 basis point at-the-money bias at held-out expiries that no in-sample statistic shows. The exponent pin is nearly immaterial by comparison, costing 6.6 basis points and moving the barrier inside Monte Carlo noise, and the construction’s guarantee holds for any admissible exponent; where the exponent’s domain does carry a guarantee, in SSVI, the data invalidates it on 92% of captures.

The panel is two weeks and there is no unsmoothed baseline; both bounds on scope are stated in Section 8.2. The code, the raw chains, the study driver and every CSV behind these tables are released, and a longer panel drops into the same pipeline unchanged.

A Reproducibility

Code, data, and the released CSVs live at

<https://github.com/siddharthqs/smile-smoothing-shootout>

and the repository README carries the exact commands, from collecting a chain snapshot to regenerating every table and figure of this paper. All computations use `RustyQLib` at commit 652e263 with the working-tree changes captured in `results/library_diff.patch`, and `results/run_manifest.json` records every filter, grid, seed and arm setting used to produce the released CSVs. Every number in the paper is computed from those CSVs by the released scripts; none is transcribed. One caveat for replication: the collector is live, so re-running the driver for the *current* session can select a different “nearest 15:30” capture — the `time` column of every row records what was used, and all archived sessions reproduce bit-exactly.

B The panel and the gate in detail

The full accounting behind Section 5.3: the cleaning cascade rule by rule (Table 9) and the fitting input it produced (Table 10); plus, because it separates nothing and requires no main-text space, the per-arm martingale check of Section 7.2.2 (Table 11). Two readings follow. First, the largest single loss (the in-the-money side of every strike) is a *design choice*, not a liquidity fact: it bounds the kept share near one half before any quote quality is consulted, and the cross-ticker spread in the kept percentage (31% for SPY down to 17% for AMD and META) is then almost entirely the spread filter acting on thinner books. Second, the tenor panel of Table 10 shows where the fitted surface is actually anchored: sub-week slices carry a median 37 pillar quotes on a span of barely $[-0.08, +0.04]$ in log-moneyness, while the 1–6M slices reach $k = -0.49$ (the moneyness window binding on the put wing) but only $k \approx +0.26$ on the call wing, where the spread cap removes the quotes. Downstream results that live in the wings (the barrier gaps most of all) inherit exactly this footprint, and every arm inherits it identically.

C Supporting figures

Figure 4 presents the admissibility rates of Table 5 in graphical form, and Figure 5 the calibration wall clocks of Table 6.

D The fitted exponent over a wider cross-section

Table 12 extends Table 4 to six further liquid single names, fitted for the exponent only. They are not part of the study panel and enter no other result in this paper; the ingest is otherwise identical, so the exponents are comparable. The exercise asks one question: where does γ sit across a wider cross-section?

Two things are visible. The exponent varies materially by name, from 0.43 to 0.64, so a pinned $\gamma = \frac{1}{2}$ is a much stronger restriction on some surfaces than on others. And the variation does not split by index against single name, which the study panel alone suggested it might. Eight of the ten single names fit above $\frac{1}{2}$, but JPM and XOM sit below it, close to DIA. The four single names of the study panel are all large-capitalisation growth names, and reading their common preference for a slower-than- $T^{-1/2}$ decay as a property of single names in general

Table 9: **Quote attrition: where the raw chain goes.** Median quote counts per capture through the shared cleaning cascade, applied once and identically upstream of all six arms. Rules in application order: unquoted, crossed or zero-bid (unq.); relative spread above 25% (spread); expiry outside 1 day–1.5 years (expiry); K/F outside $[0.6, 1.6]$ (mny.); the in-the-money side of each strike (ITM) — dropped *by design*, so the kept share is bounded near one half before liquidity is even consulted; and the residual rules (vol bounds, failed inversions, expiries left with fewer than five quotes). Failures are captures on which no expiry survives; they are recorded on every arm identically.

Ticker	Capt.	Fail	Raw	Median quotes dropped, by rule						Used	Kept %
				unq.	spread	expiry	mny.	ITM	other		
<i>Daily (15:30) slice</i>											
SPY	10	0	14 040	784	762	1 156	1 420	5 479	0	4 352	30.9
QQQ	10	0	12 086	780	1 104	873	723	4 949	0	3 650	30.0
IWM	10	0	5 710	537	635	187	282	2 456	0	1 566	27.9
DIA	10	0	4 870	370	1 015	307	130	2 032	0	971	19.5
META	10	0	7 153	712	858	717	1 607	1 895	0	1 188	17.1
AMD	10	0	6 453	648	785	631	1 718	1 559	0	1 100	17.1
TSLA	10	0	5 582	417	414	514	1 676	1 345	0	1 131	20.7
GOOGL	10	0	4 826	491	626	479	958	1 344	0	888	18.5
All daily	80	0	5 968	566	762	536	1 203	2 010	0	1 171	20.6
All intraday	1 941	59	5 910	563	785	520	1 006	1 999	0	1 158	20.4

Table 10: **What the fits received.** Top: the fitting input and the static-arbitrage repair per name (daily slice, medians per capture): expiry slices, pillar quotes per capture and per expiry, butterfly and calendar violations found in the *raw* quotes, repair adjustments applied, and the largest single vol change the repair made (absolute implied vol; 0.01 is one volatility point). Bottom: the same pillar quotes pooled by tenor bucket — where the surface is actually anchored: dense short-dated slices on narrow quoted spans, sparse long-dated ones on wide spans. Every arm fits exactly these pillars.

Per name (daily)							
Ticker	Expiries	Quotes	Quotes/expiry	Raw bfly	Raw cal.	Repairs	Max Δ vol
SPY	30	4 352	142 [105, 189]	783	105	3 453	0.117
QQQ	30	3 650	118 [99, 153]	908	88	1 621	0.206
IWM	30	1 566	45 [39, 76]	236	41	453	0.086
DIA	18	971	55 [28, 75]	258	25	479	0.091
META	20	1 188	58 [45, 70]	266	12	331	0.089
AMD	21	1 100	50 [47, 59]	279	13	372	0.138
TSLA	20	1 131	59 [40, 68]	102	13	153	0.154
GOOGL	20	888	40 [24, 69]	183	9	261	0.057
Per tenor bucket (daily, all names)							
Bucket	Slices	Quotes/expiry		Median $[k_{lo}, k_{hi}]$			
$\leq 1W$	224	37 [24, 58]		[-0.08, 0.04]			
1W–1M	510	66 [45, 99]		[-0.20, 0.09]			
1–6M	645	77 [56, 112]		[-0.49, 0.26]			
$> 6M$	526	58 [40, 108]		[-0.50, 0.45]			

Table 11: **Martingale check, per arm: all six local volatilities are usable.** Absolute relative error $|\mathbb{E}[S_T]/F_T - 1|$ in basis points and $|z|$, pooled across the 1M/3M/6M horizons; the simulation drifts at the parity forward the surface was built in (Section 4.3). No arm is separated from any other, and no cell in either slice exceeds $|z| = 3$; the only visible structure — 3M errors a couple of basis points larger than 1M and 6M — is a discretization artifact shared identically by every arm. Log-Euler with antithetic pairs, 32 768 paths, 104 steps per year, seed 20260824, identical draws across arms.

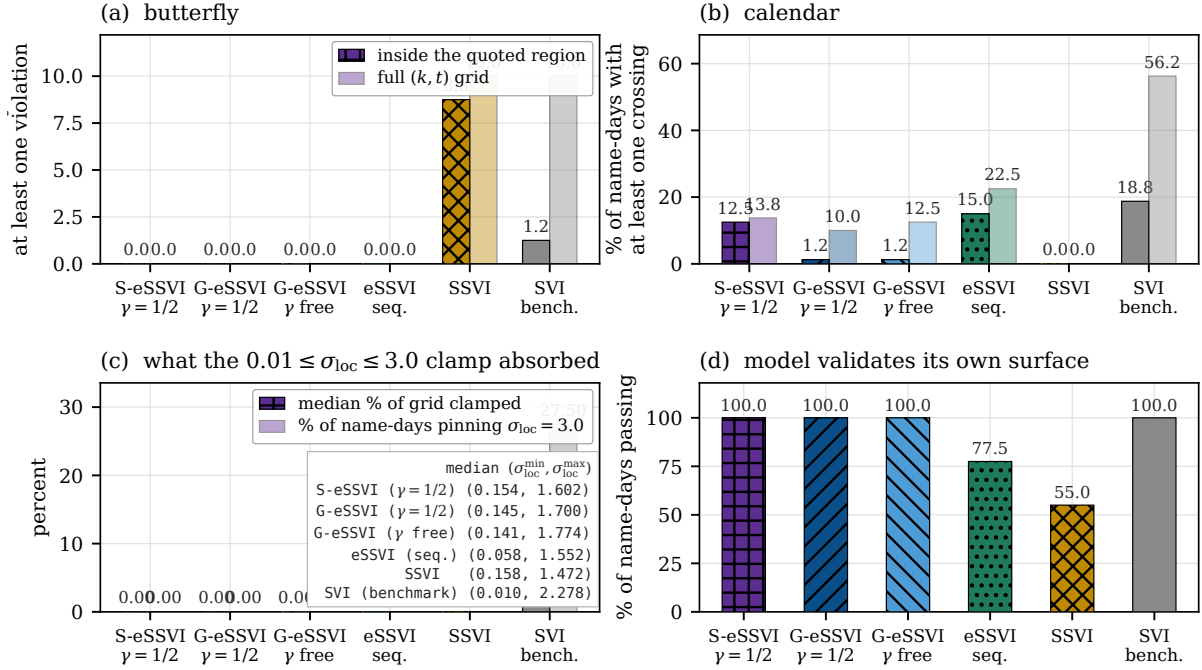
Model	Daily				Intraday			
	n	Med. bp	p99 bp	max $ z $	n	Med. bp	p99 bp	max $ z $
S-eSSVI ($\gamma=\frac{1}{2}$)	240	1.68	9.3	2.67	5 646	1.70	9.4	2.88
G-eSSVI ($\gamma=\frac{1}{2}$)	240	1.31	9.0	2.39	5 646	1.34	9.7	2.70
G-eSSVI (γ free)	240	1.30	8.9	2.41	5 646	1.34	9.6	2.68
eSSVI (seq.)	240	1.58	8.4	2.53	5 646	1.51	9.2	2.71
SSVI	240	1.66	8.3	1.70	5 646	1.45	8.4	2.02
SVI (benchmark)	240	1.25	8.9	1.85	5 646	1.23	9.4	2.30
All	1 440	1.38	9.0	2.67	33 876	1.40	9.4	2.88

Table 12: **The fitted exponent over a wider cross-section.** Median [IQR] of the fitted power-law exponent per capture on the intraday slice, and the share of captures fitting above $\frac{1}{2}$. The first eight names are the study panel and repeat Table 4; the last six are additional liquid single names fitted for the exponent only. They are not part of the study panel: no local volatility, martingale check or barrier price is computed for them, and they enter no other result in this paper. Their ingest is otherwise identical, so the exponents are comparable.

Ticker	G-eSSVI (γ free)			SSVI		
	Med.	IQR	$> \frac{1}{2}$ (%)	Med.	IQR	$> \frac{1}{2}$ (%)
<i>Index ETFs (study panel)</i>						
SPY	0.637	[0.633, 0.639]	100	0.595	[0.592, 0.596]	100
QQQ	0.509	[0.488, 0.523]	64	0.590	[0.586, 0.594]	100
IWM	0.509	[0.501, 0.524]	79	0.597	[0.591, 0.608]	100
DIA	0.464	[0.442, 0.483]	3	0.852	[0.812, 0.868]	100
<i>Single names (study panel)</i>						
META	0.583	[0.563, 0.602]	100	0.512	[0.491, 0.530]	65
AMD	0.551	[0.531, 0.577]	95	0.551	[0.515, 0.589]	83
TSLA	0.594	[0.557, 0.624]	100	0.618	[0.599, 0.634]	100
GOOGL	0.562	[0.552, 0.571]	100	0.556	[0.533, 0.573]	92
<i>Single names (exponent only)</i>						
AAPL	0.546	[0.534, 0.561]	99	0.554	[0.542, 0.567]	99
MSFT	0.548	[0.531, 0.558]	99	0.563	[0.525, 0.587]	86
AMZN	0.590	[0.578, 0.602]	100	0.614	[0.589, 0.639]	100
NVDA	0.590	[0.552, 0.606]	98	0.631	[0.618, 0.657]	100
JPM	0.429	[0.392, 0.454]	14	0.360	[0.335, 0.389]	0
XOM	0.435	[0.403, 0.468]	10	0.349	[0.310, 0.408]	6

Admissibility across the six arms

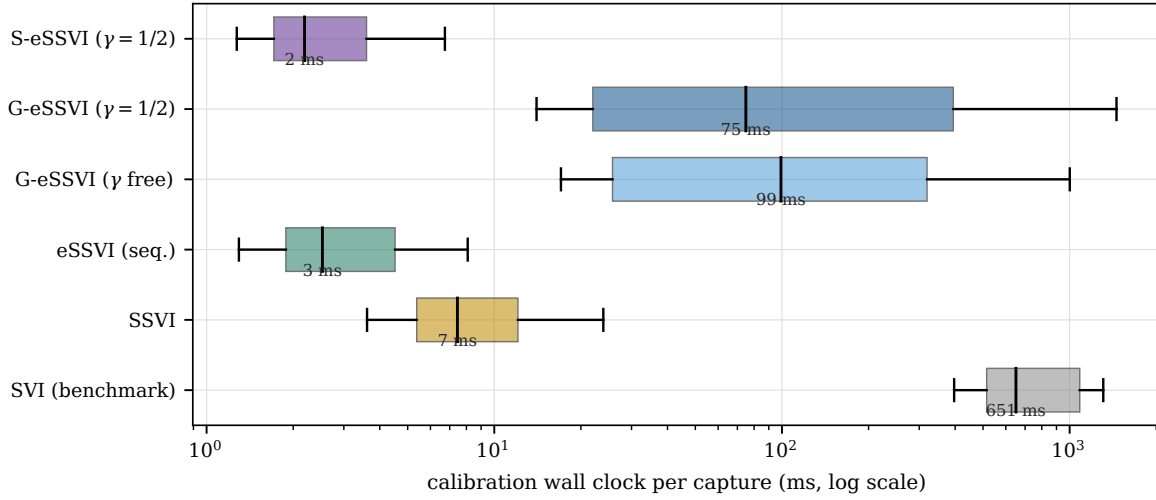
Daily slice, 80 name-days per arm. The global arms are arbitrage-free by construction at their fitted pillar slices (Mingone 2022, Prop. 3.1); anything the scans find for them lives in the interpolation between pillars, which the construction does not cover and which is measured here rather than assumed.



Counts only. min_g, the worst butterfly density on the grid, is not plotted and must not be quoted as a headline; degenerate benchmark slices give it neither a usable scale nor a meaningful mean. Whether a surface is inadmissible is well defined; how far past admissibility a degenerate fit travels is not. Clamp fractions are medians over name-days; the pinning bar counts name-days whose local vol reaches the upper clamp anywhere on the grid.

Figure 4: **Admissibility across the six arms (daily slice).** (a) Butterfly and (b) calendar violation rates, inside the quoted region and on the full (k, t) grid. The three guaranteed arms are exactly clean at their fitted pillars on every capture; their full-grid calendar entries are between-pillar interpolation, which the guarantee does not cover and the scan measures; the greedy arm's slice-local ρ path leaves it more of that than the joint solves. SSVI's inside-quotes butterfly rate, the largest in the panel, is its fitted exponent leaving the $\gamma \leq \frac{1}{2}$ sufficient-condition domain. (c) Clamp activity of the induced local volatility. (d) Each arm's own validator on its own surface: SSVI fails on every capture, for the same reason.

Calibration cost per capture, in milliseconds



Intraday slice, 1882 captures per arm, both families pooled -- the joint problem scales with the expiry count, so the global arms' boxes straddle their two regimes: median 390 ms on the ~30-expiry index chains (dim 59) against 22 ms on the single names (dim 41). Where the pin is misspecified the optimizer grinds: on SPY the pinned arm runs a median 22 iterations vs the free arm's 18.

Figure 5: **Calibration cost per capture.** Calibration wall clock on the intraday slice (boxes: IQR; whiskers: 5th–95th percentile; log axis), every fit independent, both families pooled. The global arms' boxes straddle their two regimes: a median 61-parameter problem at 307–389 ms on the index chains against 22–26 ms on the single names. The greedy S-eSSVI is the fastest arm in the panel (3.5 ms index, 1.8 ms single-name), carrying the same pillar guarantee at per-expiry cost; the sequential eSSVI runs at the same speed; the SVI benchmark, thirty per-slice fits, is the slowest arm at over a second on an index capture. Where the pin is misspecified the optimizer pays in iterations: on SPY the pinned joint arm runs 457 ms against the free arm's 361 despite one fewer parameter (see the text).

would have been an over-generalisation from that sample. Whether the split follows sector, index membership or something else is not a question two weeks of data can settle.

E Robustness to the intraday sampling time

Figure 6 repeats every share and every median on the 24-times-larger intraday slice: fit medians move by at most a few basis points, roughness medians by single-digit percentages, and the martingale gate stays at zero $|z| > 3$ cells on 33 876 rows. The statistics that move most (4–5 pp) are the ones genuinely sensitive to time of day: the between-pillar calendar rates of the joint arms (interpolation seams come and go as the short-end listings roll), the benchmark’s clamp share, and one solver-convergence flag — none of which carries a conclusion. Nothing that does carry one changes sign or order.

F The eSSVI construction in detail

This appendix collects the parts of the construction of Section 3.4 that a reader reproducing the calibration needs and a reader following the results does not: the correction to the calendar condition, the mechanics of Mingone’s recursion, and the feasibility argument behind the greedy pass.

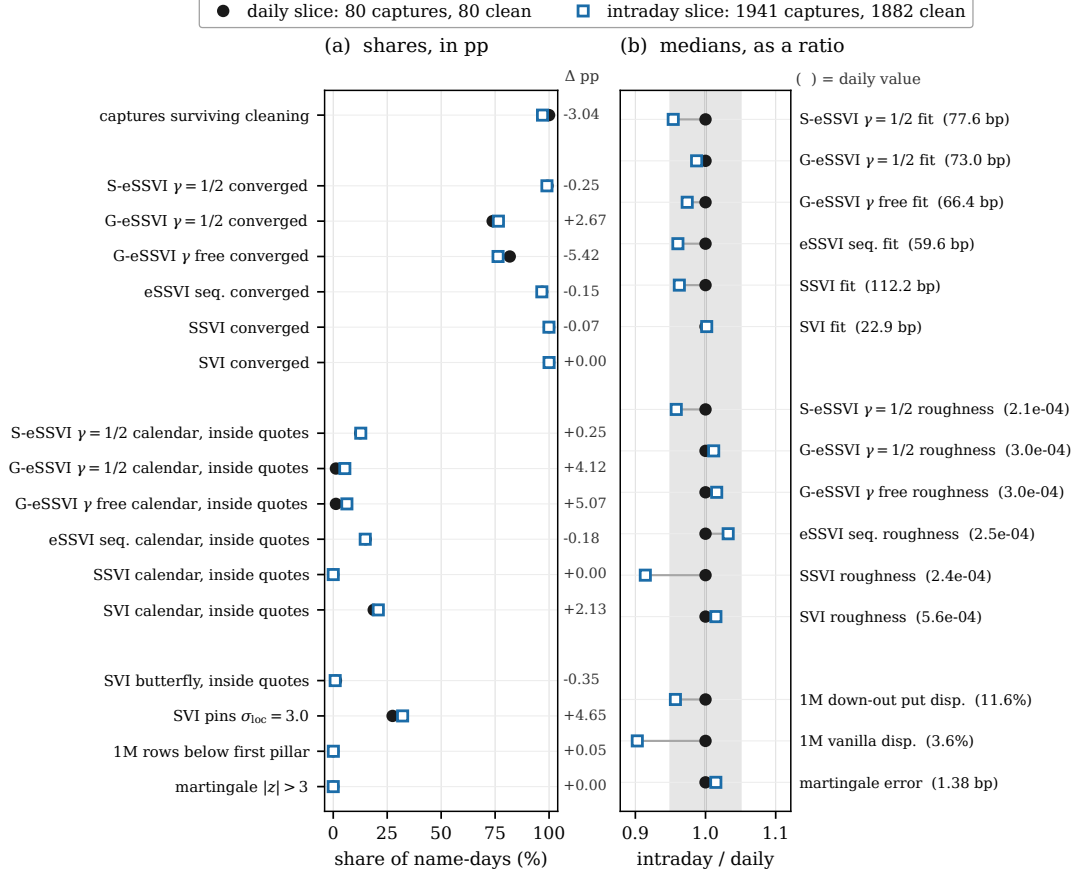
The corrected calendar condition. The calendar-spread part of (14) is due to Hendriks and Martini (2019) in spirit but not in letter. Their Proposition 3.1 stated the no-crossing condition as a two-sided *squared* inequality; Pasquazzi (2023) showed that proposition to be incorrect — slices satisfying it can still cross — and proved the sharp characterization. For $\theta_2 > \theta_1$, two eSSVI slices do not cross if and only if $\psi_2 \leq \psi_1 \theta_2 / \theta_1$ (or the weaker second branch of his condition S) holds jointly with $\psi_2(1 \pm \rho_2) \geq \psi_1(1 \pm \rho_1)$; the latter says the two asymptotic wing slopes of total variance must not decrease in maturity. The latter pair is algebraically the max-ratio lower bound $\psi_1 p_2 \leq \psi_2$ of (14). The conditions in the form used throughout this paper — and by the calibration lineage we build on (Corbetta et al., 2019; Mingone, 2022) — are therefore Pasquazzi’s *corrected* sufficient set, and are untouched by the error; his tangency cases arise only when ψ/θ increases in maturity, which (14) excludes. Our dense-grid calendar scans (Section 7.1.4) verify the same conclusion numerically on every fitted surface.

The recursion, and why the tube is never empty. In (15) the free parameters are $\rho_1, \dots, \rho_N \in (-1, 1)$, $\theta_1 > 0$ and $a_2, \dots, a_N > 0$. Two features do the work. The backward recursion $S_i = \min(f_i, S_{i+1}/p_{i+1})$ propagates every later slice’s butterfly cap back through the wing-slope ratios, so the ceiling C_{ψ_i} already accounts for slices not yet chosen; this is the look-ahead that makes the problem joint. The multiplicative p_i in $\theta_i = \theta_{i-1} p_i + a_i$ is what keeps the tube non-empty: because the floor is $A_{\psi_i} = \psi_{i-1} p_i$ and the calendar ceiling is $\psi_{i-1} \theta_i / \theta_{i-1}$, forcing θ to grow by at least the factor p_i guarantees the ceiling clears the floor whatever ρ_i is chosen. Our implementation verifies Proposition 3.1 mechanically over 500 randomized parameter draws as a unit test.

Feasibility of the greedy pass. The greedy arm restricts ρ_i to the values at which slice i ’s tube is non-empty, which requires $p_i \leq \theta_i / \theta_{i-1}$ on the calendar side and $\psi_{i-1} p_i < f_i(\rho_i)$ on the butterfly side; the second binds where dense short-dated listings make $f_i \sim \sqrt{4\theta_i}$ small.

Nothing here is an artefact of the 15:30 cut

24 times as many captures, computed through the identical pipeline: every share moves by at most 5.4 pp and every median by at most 9.7%. Shaded band in (b): $\pm 5\%$.



Martingale tails: 0.00% of rows at $|z| > 3$ daily ($n = 1440$) against 0.00% intraday ($n = 33876$), both below the 0.27% expected under normality. The collector runs live, so the capture nearest 15:30 in the current session can move between runs; everything else here reproduces bit-exactly. Barrier dispersions are the five-arm menu on calibrated horizons, matching fig. 3.

Figure 6: **The 15:30 cut against the full intraday panel.** Every share statistic compared in percentage points (left) and every median as an intraday/daily ratio against a $\pm 5\%$ band (right). The largest movers are statistics genuinely sensitive to time of day — between-pillar calendar rates for the joint arms and the benchmark’s clamp share — none of which carries a conclusion.

Both hold strictly at $\rho_i = \rho_{i-1}$, since p_i is then 1, θ has strictly increased, and ψ_{i-1} sits strictly below a butterfly cap that only grows with θ . Both also tighten monotonically as ρ_i moves away from ρ_{i-1} in either direction. The feasible interval therefore always contains ρ_{i-1} and is never empty, and each edge is found by bisection outward from ρ_{i-1} . One degenerate case is handled explicitly: when two pillars carry near-equal ATM variance the interval collapses to a point, and the previous slice's ρ is held rather than fitted.

G Formulas as implemented

For the referee's convenience, the exact quantities computed.

Butterfly function and Dupire. $g(k)$ is evaluated from (3) using closed-form $w'(k)$ and $w''(k)$ for every arm (the three guaranteed arms share the eSSVI slice form). For SVI, with $d = k - m$ and $r = \sqrt{d^2 + \sigma^2}$: $w = a + b(\rho d + r)$, $w' = b(\rho + d/r)$, $w'' = b\sigma^2/r^3$. For SSVI and eSSVI, with φ the curvature ($\varphi = \psi/\theta$ in the eSSVI case), $u = \varphi k + \rho$ and $R = \sqrt{u^2 + 1 - \rho^2}$: $w = \frac{\theta}{2}(1 + \rho\varphi k + R)$, $w' = \frac{\theta}{2}\varphi(\rho + u/R)$, $w'' = \frac{\theta}{2}\varphi^2(1 - \rho^2)/R^3$. The Dupire numerator $\partial_t w$ is taken from the same interpolation rule used to build the surface (Section 4), and σ_{loc}^2 from (6) with the guards of Section 4.

Roughness. The reported roughness is the mean absolute second difference of σ_{loc} along k at spacing $\Delta k = 0.01$, averaged over the published (k, t) grid: $\text{mean } |\sigma_{\text{loc}}(k + \Delta k, t) - 2\sigma_{\text{loc}}(k, t) + \sigma_{\text{loc}}(k - \Delta k, t)|$. Dividing by Δk^2 rescales but does not reorder; the raw second difference is what the CSV column `roughness_mean_abs_d2` contains, and Table 7 reports it $\times 10^4$.

Dispersion. For each (name-day, maturity, product, barrier level) the arms of the practical menu (the three guaranteed eSSVI arms, sequential eSSVI, SSVI) are priced on one shared path set and the dispersion is (max – min) over the menu's prices, reported as a fraction of their mean, in basis points of spot, and divided by the mean Monte Carlo standard error of those prices.

ATM level and skew. The stability instrument reads $\sigma_{\text{ATM}} = \sqrt{w(0)/t}$ and $\partial_k \sigma|_{k=0} = w'(0)/(2\sqrt{w(0)t})$ in closed form from the fitted parameters, at the fitted pillar nearest one month. Because every arm fits the same repaired pillars, the selected expiry is identical across arms at every snapshot, so an expiry roll moves all the series together and cannot masquerade as model instability.

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